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PHYSICS

1) A man holding a flag is running in north east direction with speed 10 m/s. Wind is blowing in east direction with speed $5\sqrt{2}$ m/s. Find the direction in which flag will flutter?

(1) South west
(2) South East
(3) South
(4) East

2) A boat moves along the flow of river between two fixed point A and B. It takes time 3 min when going down stream and takes 6 min when going upstream between these two point. What time it will take in still water to cover the distance equal to AB.

(1) 2 min
(2) 3 min
(3) 4 min
(4) 5 min

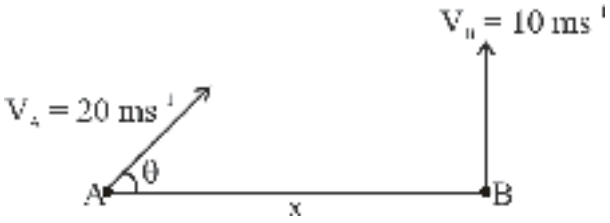
3) A man is running on level road with speed 4m/s. He feels rain drops are falling vertically downwards. If speed of rain drops are 8 m/s wrt. ground. Find out speed of rain drops wrt man.

(1) 4 m/s
(2) $4/\sqrt{3}$m/s
(3) $4\sqrt{3}$m/s
(4) $2\sqrt{3}$m/s

4) A car is moving horizontally along a straight line with a uniform velocity of 25 m s^{-1}. A projectile is to be fired vertically from this car in such a way that it will return to it after it has moved 100 m. The speed of the projection w.r.t car must be : (g = 9.8 m/s^2)

(1) 9.8 m s^{-1}
(2) 19.6 m s^{-1}
(3) 15.6 m s^{-1}
(4) 24.6 m s^{-1}

5) Two particles A and B are projected simultaneously in direction shown in figure with velocities V_A = 20 ms^{-1} and V_B = 10 ms^{-1} respectively. They collide in air after $\frac{1}{2}$ sec. Find angle θ and Distance x



(1) $60°, 10\sqrt{3}\,m$

(2) $45°, 3\sqrt{3}\,m$

(3) $15°, 6\sqrt{3}\,m$

(4) $30°, 5\sqrt{3}\,m$

6) Car A is 100 m towards North of car B. Car A starts moving towards east with speed 10 m/s simultaneously car B starts to move towards north with same speed find out time when seperation between the cars become minimum.

(1) 5 s

(2) $\frac{5}{\sqrt{2}}s$

(3) $5\sqrt{2}s$

(4) $\frac{5}{2}s$

7) A stone is thrown upwards from a tower with a velocity 50 m/s another stone is simultaneously thrown downwards from the same location with a velocity 50 m/s. When the first stone is at the highest point, the relative velocity of the second stone with respect to the first stone is (assume that second stone has not yet reached the ground).

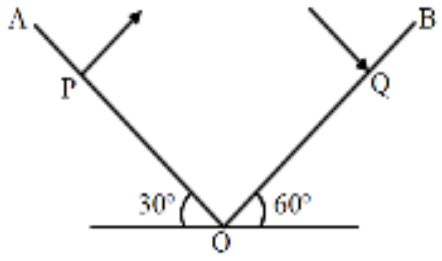
(1) 50 m/s

(2) 60 m/s

(3) + 200 m/s

(4) – 100 m/s

8) Two inclined are arranged as shown. A particle is thrown penpendicular to plane OA with velocity $10\sqrt{3}\,m/sec$ & strikes perpendicular to OB. find velocity at point Q.



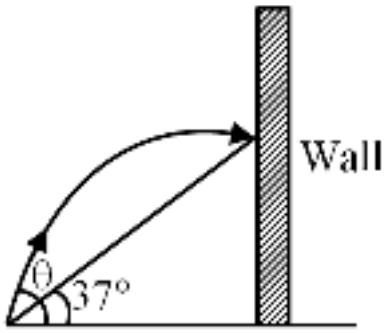
(1) 10 m/ sec

(2) $5\sqrt{3}\,m/sec$

(3) $\frac{10}{\sqrt{3}}m/sec$

(4) $\frac{5}{\sqrt{3}}$m/sec

9) A shot is fired at an angle θ to the horizontal such that it strikes the wall while moving horizontally. Find the initial angle of projection θ :-



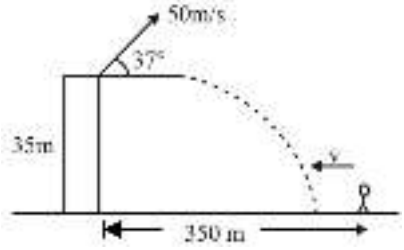
(1) $\theta = \tan^{-1}\left(\frac{2}{5}\right)$

(2) $\theta = \tan^{-1}\left(\frac{3}{8}\right)$

(3) $\theta = \tan^{-1}\left(\frac{3}{2}\right)$

(4) $\theta = 53^{\circ}$

10) A particle is projected from a tower with speed 50 m/s at angle 37° with horizontal. Find out minimum velocity of man such that he will catch the ball. Distance between man and building is 350 m and height of tower is 35 m.



(1) 20 m/s

(2) 25 m/s

(3) 30 m/s

(4) 10 m/s

11) A ball is thrown from the roof of a building of height 44 m with speed V_0 at an angle θ below the horizontal. It lands 2 seconds later at a point 30 m from the base of the building. If $\tan\theta = \frac{x}{10}$ then, find the value of x ($g = 10m/s^2$)

(1) 8

(2) 6

(3) 4

(4) 5

12) A particle is thrown horizontally from top of a tower with speed 30 m/s. It reaches to ground in 4s. Find out elevation angle of top of the tower at landing position.

(1) 37°

(2) 53°

(3) $\tan^{-1}\left(\frac{1}{3}\right)$

(4) $\tan^{-1}\left(\frac{2}{3}\right)$

13) A ball rolls off the top of a staircase with a horizontal velocity u m/s. If the steps are h metre high and b metre wide, the ball will hit the edge of the nth step, if:

(1) $n = \frac{2hu}{gb^2}$

(2) $n = \frac{2hu^2}{gb}$

(3) $n = \frac{2hu^2}{gb^2}$

(4) $n = \frac{hu^2}{gb^2}$

14) A projectile is given initial velocity of $\left(\hat{i} + 2\hat{j}\right)$ m/s. The equation of the path is : (g = 10 m/s^2)

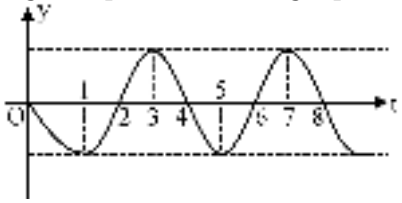
(1) $y = 2x - 5x^2$

(2) $y = x - 5x^2$

(3) $4y = 2x - 5x^2$

(4) $y = 2x - 25x^2$

15) For the given position time graph match the following and choose the correct option. (Graph is for SHM)



Column - I		Column - II	
(a)	velocity at = 1 sec.	(P)	Positive
(b)	acceleration at t = 1 sec.	(Q)	Negative
(c)	position at t = 5 sec.	(R)	Zero
(d)	velocity at t = 8 sec.		

	(a)	(b)	(c)	(d)
(1)	P	Q	R	P
(2)	R	P	Q	R
(3)	R	R	Q	P
(4)	R	P	Q	Q

(1) 1

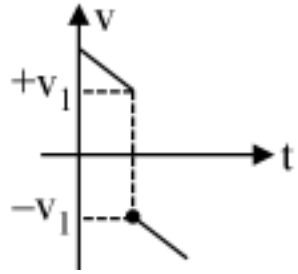
(2) 2

(3) 3

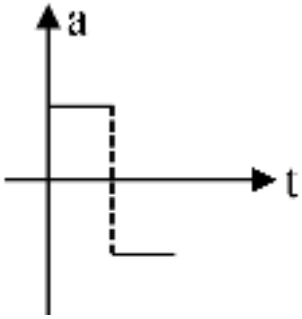
(4) 4

16) A ball is thrown upward in a room. The ball rises up, hits the ceiling and fall down. Which of the following graph represent the motion of ball.

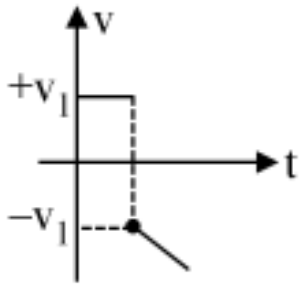
(1)



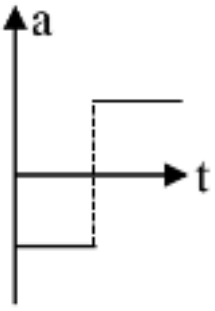
(2)



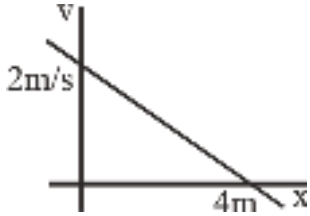
(3)

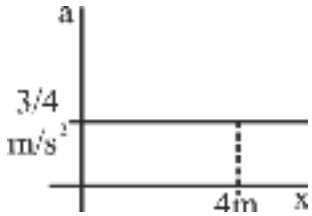


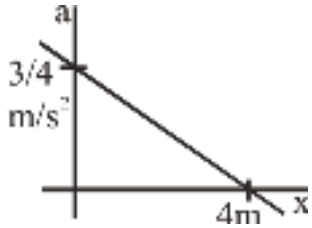
(4)

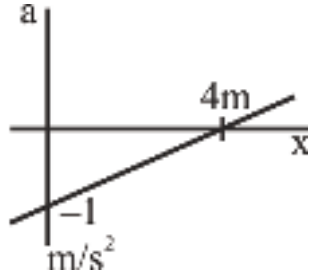


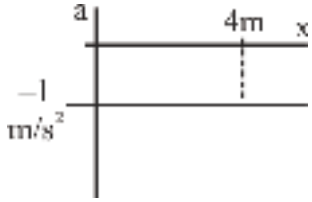
17) For a particle travelling on x-axis the velocity position curve is given. Its acceleration position

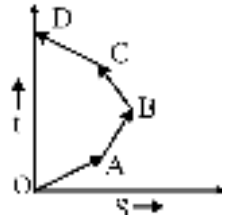
curve will be :- 

(1) 

(2) 

(3) 

(4) 

18) Which of the following options is correct for the object having a straight line motion represented by the following graph :- 

(1) The object moves with constantly increasing velocity from O to A and then it moves with constant velocity.

(2) Velocity of the object increases uniformly

(3) Average velocity is zero

(4) The graph shown is impossible

19) A paratrooper after bailing out falls 50 m without friction. When parachute opens, it decelerates at 2 m/s^2. He reaches the ground with a speed of 3 m/s. At what height did he bail out :

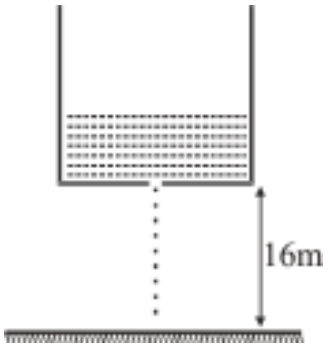
(1) 293 m

(2) 111 m

(3) 91 m

(4) 182 m

20) Water drops fall with negligible velocity at regular intervals from a hole at the bottom of a vessel placed 16 m from the ground. The ninth drop is about to fall when the first drop just falls on the floor. Find the distance between the third and fifth drop at this instant in meters.



(1) 2

(2) 5

(3) 8

(4) 9

21) A body starts from rest from the origin with an acceleration of 6 m/s^2 along the x-axis and 8 m/s^2 along the y-axis. Its distance from the origin after 4 seconds will be :-

(1) 56 m

(2) 64 m

(3) 80 m

(4) 128 m

22) Velocity of a particle changes with displacement as $V = (3x^2 + 4)$ m/s, then acceleration of particle at x = 2m is.

(1) 192 m/sec^2

(2) 16 m/sec^2

(3) 12 m/sec^2

(4) 150 m/sec^2

23) A particle moves rectilinearly. its displacement x at time t is given by $x^2 = t^2 + 1$ then its acceleration at time 't' is :

(1) $\frac{1}{x^3}$

(2) $\frac{1}{x} - \frac{1}{x^2}$

(3) $-\frac{t}{x^2}$

(4) $-\frac{t^2}{x^3}$

24) A car moves a distance of 60 km with 10 km/hr and next 120 km with 20 km/hr. Calculate average speed for total journey :-

(1) 15 km/hr
(2) 12 km/hr
(3) 18 km/hr
(4) 16 km/hr

25) A person moves 30 m due north, then 20 m due east then $20\sqrt{2}\,m$ due south-west. Find net displacement.

(1) 10 m due N
(2) 20 m due S
(3) 70 m due E
(4) 0

26) A charge Q is placed at each of the opposite corners of a square. A charge q is placed at each of the other two corner. If the net electrical force on Q is zero, then Q/q equals :-

(1) $-2\sqrt{2}$
(2) –1
(3) 1
(4) $-\frac{1}{\sqrt{2}}$

27) A particle of mass 10^{-3} kg and charge 5μC is projected against an uniform electric field 2×10^{5} N/C at a speed of 20 m/s. The distance travelled by it before coming to rest is :-

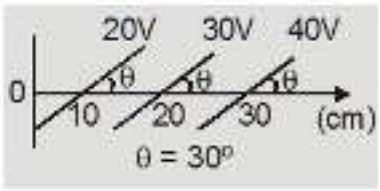
(1) 0.1 m
(2) 0.2 m
(3) 0.3 m
(4) 0.4 m

28) Electric potential in a region is $V=\frac{3x^2}{2}-\frac{y^2}{4}$, where x and y are in metre and V is in volt electric field intensity (in V/m) at a point (1m, 2m) is :-

(1) $3\hat{i}-\hat{j}$
(2) $-3\hat{i}+\hat{j}$
(3) $6\hat{i}-2\hat{j}$
(4) $-6\hat{i}+2\hat{j}$

29) Some equipotential surfaces are shown in the figure. The magnitude and direction of the electric

field is-



(1) 100 V/m making angle 120° with the x-axis

(2) 100 V/m making angle 60° with the x-axis

(3) 200 V/m making angle 120° with the x-axis

(4) none of the above

30) Two point P and Q are maintained at the potentials of 10V and –4V respectively. The work done in moving 100 electrons from P to Q is :-

(1) -19×10^{-17} joule

(2) 9.60×10^{-17} joule

(3) -2.24×10^{-16} joule

(4) 2.24×10^{-16} joule

31) **Assertion :** The accelerations of electron and proton are different in same electric field.
Reason : Electric force acting on unit positive charge is independent of mass.

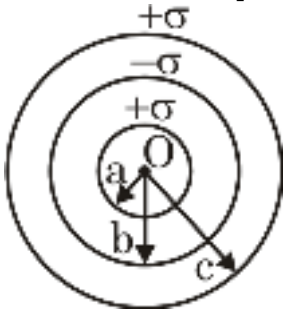
(1) Both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

(2) Both Assertion and Reason are true, but Reason is not correct explanation of the Assertion.

(3) Assertion is true, but the Reason is false.

(4) Both Assertion and Reason are false.

32) Electrostatic potential at the common centre O of the system shown in figure, is :-



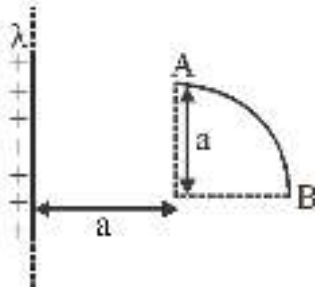
(1) $\frac{\sigma}{\epsilon_0}\left(\frac{1}{a}-\frac{1}{b}+\frac{1}{c}\right)$

(2) $\frac{\sigma}{\epsilon_0}(a-b+c)$

(3) $\frac{\sigma}{2\epsilon_0}\left(\frac{1}{a}-\frac{1}{b}+\frac{1}{c}\right)$

(4) $\frac{\sigma}{2\epsilon_0}(a-b+c)$

33) Work done in shifting a point charge q from A to B along a circular arc as shown, will be



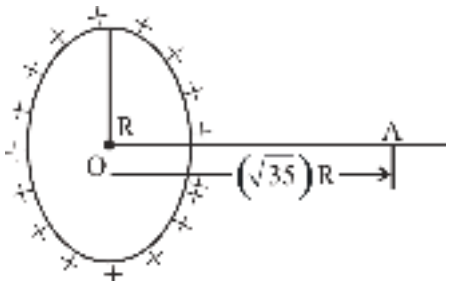
(Consider infinity line charge of density λ and arc AB are in same plane) :-

(1) 2kλq□n2

(2) –2kλq□n2

(3) kλq□n2

(4) –kλq□n2



34) A ring has total charge Q. Find work done by external agent to move –Q charge from point A to centre of ring :-

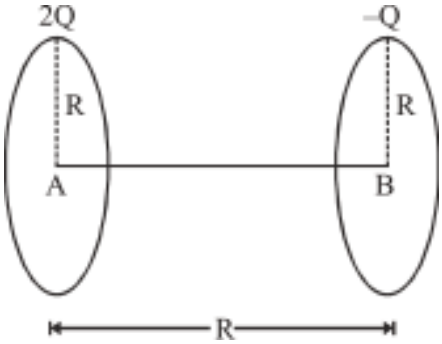
(1) $\frac{5KQ^2}{6R}$

(2) $\frac{-5}{6}\frac{kQ^2}{R}$

(3) $\frac{3}{2}\frac{kQ^2}{R}$

(4) $\frac{kQ^2}{4R}$

35) In the following arrangement. Choose correct option:-



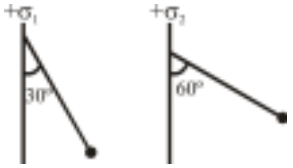
(1) $V_A = \frac{Q(2\sqrt{2}-1)}{4\pi\in_0\sqrt{2}R}$

(2) $V_B = \frac{Q(\sqrt{2}-1)}{4\pi\in_0 R}$

(3) W.D. in moving a charge +q from B to A $W_{BA} = \frac{qQ(3\sqrt{2}-3)}{4\sqrt{2}\pi\in_0 R}$

(4) all options are correct

36) The angles of inclination of the suspended charged pendulum are 30º & 60º due to electric field of the infinite sheets of surface charge densities σ_1 and σ_2 respectively, then σ_1/σ_2



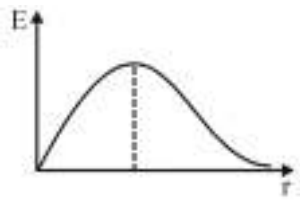
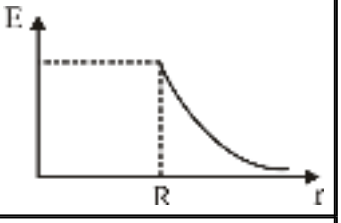
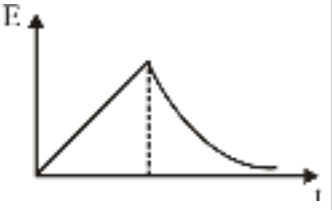
(1) $\sqrt{3}$

(2) 3

(3) $1/\sqrt{3}$

(4) 1/3

37) Column II corresponds to the graph of magnitude of electric field versus distance from centre of charge distribution in column I

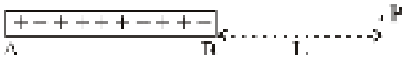
Column-I		**Column-II**	
(A)	Ring along its axis	(P)	
(B)	Uniformly charged solid sphere	(Q)	
(C)	Spherical conducting shell	(R)	

(1) A→P, B→Q, C→R

(2) A→P, B→R, C→Q

(3) A→R, B→P, C→Q

(4) A→Q, B→P, C→R

38) A rod AB of length L has charge density $\lambda = \alpha (2L - x)^2$ where x is distance from end A and α is a constant then electric field at point P will be :- 

(1) K α L

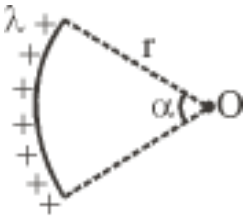
(2) $\frac{K\alpha}{L}$

(3) $K\alpha L^2$

(4) $\frac{K\alpha}{L^2}$

39)

If electric field at centre of charged arc of radius r and linear charge density λ is $\frac{\sqrt{3}\lambda}{4\pi\in_0 r}$. Then α is :



(1) 60°

(2) 120°

(3) 30°

(4) 135°

40) Electric field is maximum due to a charged ring of radius R at a distance from circumference of the ring :

(1) $\frac{R}{\sqrt{2}}$

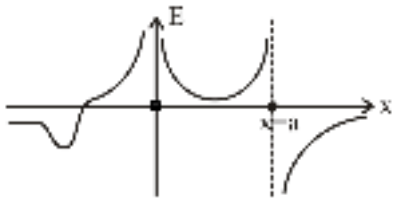
(2) $\sqrt{\frac{3}{2}}R$

(3) $\frac{\sqrt{3}}{2}R$

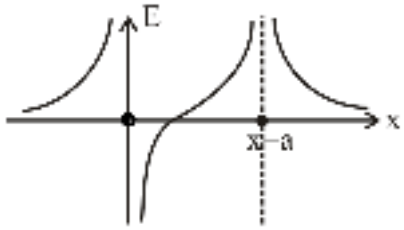
(4) $\sqrt{\frac{3}{2}R}$

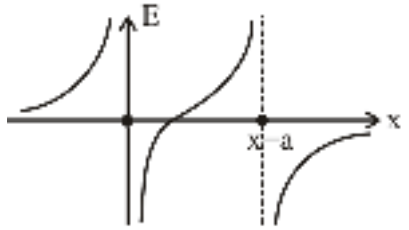
41) Two point charges –Q and –2Q are placed at x = 0 and x = +a respectively which graph represent the variation of electric field along x axis :

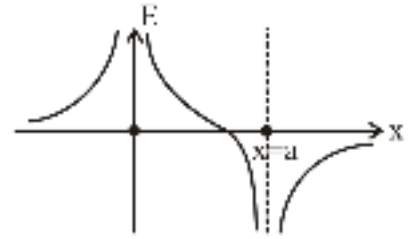
(1)



(2)



(3) 

(4) 

42) An infinite number of charges, each of charge 1 µC, are placed on the x-axis with co-ordinates x = 1, 2, 4, 8, ...∞. If a charge of 1 C is kept at the origin, then what is the net force acting on 1 C charge

(1) 9000 N

(2) 12000 N

(3) 24000 N

(4) 36000 N

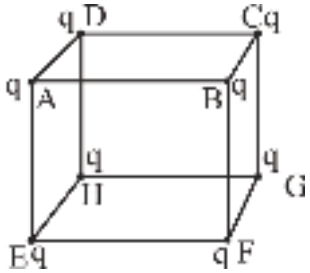
43) A charged gold leaf electroscope having enclosed air has its leaves apart by a certain amount of, when the electroscope is subjected to x - rays then the leaves -

(1) Further dilute

(2) Start oscillating

(3) Collapse

(4) Remains some



44) Eight point charges having magnitude q are fixed at vertices of a cube.

Statement 1 : electric flux through square surface ABCD of the cube is $\frac{q}{24\epsilon_0}$

Statement 2 : electric flux through cube is $\frac{q}{\epsilon_0}$

Statement 3 : electric flux through square surface EFGH of the cube is $\frac{q}{6\epsilon_0}$

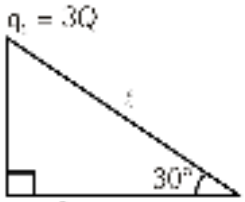
Which of the following statements are correct

(1) 1 & 2

(2) 2 & 3

(3) 1 & 3

(4) All



45) In the given fig. If $q_3 = Q$ $q_2 = 2Q$ F_1 = Magnitude of Force on q_1 due to q_2
F_2 = Magnitude of Force on q_1 due to q_3

Find $\frac{F_1}{F_2}$

(1) $\frac{7}{2}$

(2) $\frac{3}{2}$

(3) $\frac{3\sqrt{3}}{2}$

(4) $\frac{9}{2}$

CHEMISTRY

1) According to VBT, which of the following overlapping results pi-bond in O_2 molecule formation, when z-axis is internuclear axis ? I. 2s – 2s II. $2p_x - 2p_x$
III. 1s – 1s IV. $2p_y - 2p_y$ V. $2p_z - 2p_z$

(1) II,IV,V

(2) II, V

(3) II, IV

(4) IV, V

2) Nitrogen exists in form of N_2 but phosphorous does not exists in form of P_2 the reason is -

(1) Triple bond present between phosphorous atom.

(2) pπ – pπ bonding is weak between P-atoms.

(3) pπ – pπ bonding is strong between P-atoms.

(4) Multiple bond form easily between P-atoms.

3) The no . of excited state of centre atom to the formation of IF_7, PCl_5, SO_2Cl_2 and NH_3 are X,Y,Z and W respectively, the correct order of :

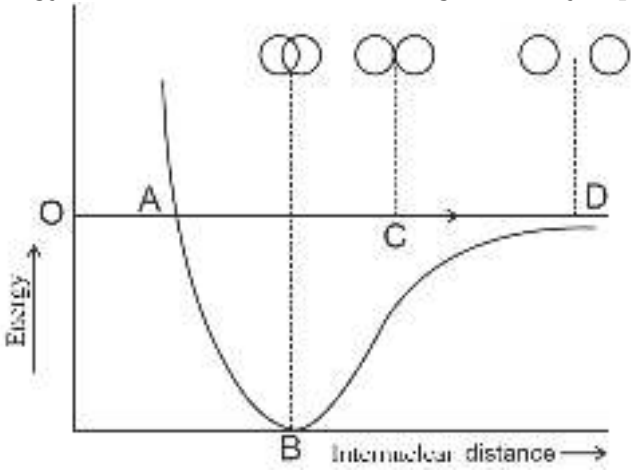
(1) X > Z > Y > W

(2) X = Z = Y = W

(3) X > Z = Y < W

(4) X < Z < Y < W

4) Look at the following potential energy curve which of the following correctly represents the most stable state of hydrogen molecule.



(1) A

(2) C

(3) B

(4) D

5) Which compound dose not exist :

(1) CF_6^{-2}

(2) BF_6^{-3}

(3) F_3^{-1}

(4) All of these

6) Choose incorrect option regarding Lewis dot structure ?

(1) The Lewis dot structure provide a picture of bonding in molecules and ions in terms of the shared pair electrons and the octet rule.

(2) The lowest energy structure is one with the smallest formal charge on the atoms.

(3) The lowest energy structure is one with the highest formal charge on the atoms.

(4) The Lewis dot structure of CO_2 has equal no. of bond pair & lone pair.

7) Which of the following not follow octet rule :

(1) B_2H_6

(2) $AlCl_3$

(3) NaF

(4) Both (1) and (2)

8) Which of the following species is neither hypervalent nor hypovalent ?

(1) ClO_4^-

(2) BF_3

(3) SO_4^{2-}

(4) CO_3^{2-}

9)

(i) Process of adding electron to an atom may be exothermic or endothermic.
(ii) In most of the elements energy is released when an electron is added to the atom.
(iii) Halogens have maximum ΔH_{eg} in their respective period because of their stable electronic configuration.
(iv) Electron affinity order S < O.
(v) ΔH_{eg} is maximum positive for inert gases in their respective period.
Select the correct code in terms of true (T) and false (F).

(1) TFFTF
(2) FTTFT
(3) TTFFT
(4) FFFTF

10) The correct order of acidic strength is :-

(1) $Cl_2O_7 > SO_3 > P_4O_{10}$
(2) $CO_2 > N_2O_5 > SO_3$
(3) $Na_2O > MgO > Al_2O_3$
(4) $K_2O > CaO > MgO$

11) **Statement 1** : Na^+ and Al^{3+} are isoelectronic but the magnitude of ionic radius of Al^{3+} is less than that of Na^+
Statement 2 : The magnitude of effective nuclear charge of the outer shell electrons in Al^{3+} is greater than that in Na^+
Which of the following is correct for these is statements :

(1) Statement-1 is True, Statement-2 is true, Statements-2 is a correct explanation for Statement-1
(2) Statement-1 is true, Statement-2 is true, Statements-2 is NOT a correct explanation for Statement-1
(3) Statement-1 is True, Statement-2 is False
(4) Statement-1 is False, Statement-2 is True

12) Which is correct :-

	Property		**KJ mol^{-1}**
(a)	EA of F	(p)	349
(b)	EA of Cl	(q)	1256
(c)	IP of F	(r)	328
(d)	IP of Cl	(s)	1680

Correct match is

(1) a = p, b = r, c = q, d = s

(2) a = r, b = q, c = s, d = p

(3) a = r, b = q, c = p, d = s

(4) a = r, b = p, c = s, d = q

13) The five successive ionization enthalpies of an element are 800,2427, 3658,25024 and 32824 kJ mol^{-1}, The number of valence electrons in the element is:

(1) 2

(2) 3

(3) 4

(4) 5

14) Which of the following show maximum ionisation energy :

(1) F

(2) Ne

(3) $\overset{\oplus}{Ne}$

(4) $\overset{\ominus}{F}$

15) Which of the following statement is **incorrect** ?

(1) The second ionisation enthalpy of oxygen is greater than that of flourine element.

(2) The third ionization enthalpy of phosphorus is greater than that of aluminium.

(3) The first ionization enthalpy of aluminium is slight greater than that of gallium.

(4) The second ionisation enthalpy of copper is greater than that of zinc.

16) If a substance with half-life 3 days is taken to another place in 12 days. What part of initial amount of substance is left now :-

(1) 1/4

(2) 1/8

(3) 1/16

(4) 1/32

17) Rate of formation of SO_3 in the following reaction $2SO_2 + O_2 \rightarrow 2SO_3$ is 100 gm/min. Rate of disappearance of O_2 is:

(1) 50 gm/min

(2) 200 gm/min

(3) 40 gm/min

(4) 20 gm/min

18) A certain reaction obeys the rate equation (in the integrated form) $(n-1)kt = \left[C^{(1-n)} - C_0^{(1-n)}\right]$ where C_0 is the initial concentration and C_t is the concentration after time t. What is the unit of k for n = 3 :-

(1) sec^{-1}

(2) litre mol^{-1} sec^{-1}

(3) mol $litre^{-1}$ sec^{-1}

(4) $litre^{2}$ mol^{-2} sec^{-1}

19) The half-life of a first order reaction is one hour. During what time interval will the concentration be reduced to $\frac{7}{8}$ of its initial value :-

(1) 6.32 minutes

(2) 17.4 minutes

(3) 23.2 minutes

(4) 11.6 minutes

20) For $N_2 + 3H_2 \rightarrow 2NH_3 + 22$ kcal, E_a (energy of activation) is 70 kcal. Hence, E_a of $2NH_3 \rightarrow N_2 + 3H_2$ is :-

(1) 92 kcal

(2) 70 kcal

(3) – 70 kcal

(4) – 92 kcal

21) The rate of reaction increases on the adsorption on catalyst. Because.....

(1) Adsorption decreases E_a.

(2) Concentration of reactants increases on the active sites of catalyst.

(3) Adsorption energy is produced which increase rate of reaction.

(4) All of these

22) Consider the following statements with respect of zero order reaction
(i) The rate of the reaction is independent of reactant concentration
(ii) The rate of the reaction is independent of temperature
(iii) The rate constant of the reaction is independent of temperature
(iv) The rate constant of the reaction is independent of reactant concentration
Choose the correct statement(s).

(1) (i) only

(2) (i) and (ii) only

(3) (iii) and (iv) only

(4) (i) and (iv) only

23) If $t_{1/2} \times \sqrt{a}$ = constant, then order of reaction is (a = initial concentration)?

(1) $\frac{1}{2}$

(2) $-1/2$

(3) $\frac{3}{2}$

(4) $-\frac{3}{2}$

24)

K	Temp.
2×10^{-6}	300 K
?	600 K

If $E_a = 0$, then rate constant at 600 K is :-

(1) 4×10^{-6}

(2) 1×10^{-6}

(3) 2×10^{-6}

(4) 3×10^{-6}

25) $A \rightarrow 2B$; E_{a_1}

$\frac{1}{2} A \rightarrow B$; E_{a_2}

Correct relation between E_{a_1} and E_{a_2} is :-

(1) $E_{a_1} = 2E_{a_2}$

(2) $E_{a_1} = \frac{E_{a_2}}{2}$

(3) $E_{a_1} = E_{a_2}$

(4) $E_{a_1} = 4E_{a_2}$

26) The hydrogenation of vegetable ghee at 25°C reduces the pressure of H_2 from 2 atm to 1.2 atm in 50 min. Calculate the rate of reaction in terms of change of Molarity per second :-

(1) 3.27×10^{-2} Ms^{-1}

(2) 2.09×10^{-6} Ms^{-1}

(3) 1.09×10^{-5} Ms^{-1}

(4) 10.9×10^{-5} Ms^{-1}

27)

In a pseudo first order hydrolysis of ester in water, the following results were obtained :-

t/s	0	30	60	90
Ester/mol L^{1}	0.55	0.31	0.17	0.085

What will be the average rate of reaction between the time interval 30 to 60 seconds ?

(1) $1.91 \times 10^{-2}\ s^{-1}$

(2) $4.67 \times 10^{-3}\ mol\ L^{-1}\ s^{-1}$

(3) $1.98 \times 10^{-3}\ s^{-1}$

(4) $2.07 \times 10^{-2}\ s^{-1}$

28) In the reaction $2A \rightarrow 3B$, initially 5 moles of A are taken, after 10 min 6 mol of B is formed. The average rate of disappearance of A will be

(1) $\frac{1}{10}\frac{mol}{min}$

(2) $\frac{4}{10}\frac{mol}{min}$

(3) $\frac{5}{10}\frac{mol}{min}$

(4) $\frac{6}{10}\frac{mol}{min}$

29)

The half life period of a first order chemical reaction is 6.93 minutes. The time required for the completion of 99% of the chemical reaction will be (log 2 = 0.301) :-

(1) 46.06 minutes

(2) 460.6 minutes

(3) 230.3 minutes

(4) 23.03 minutes

30) For a zero order reaction $A \rightarrow B$, the reaction rate at reactant concentration of 0.01 M is found to be 2.0×10^{-5} M sec^{-1}. The half life period of the reaction is :-

(1) 333 sec.

(2) 300 sec.

(3) 346.5 sec.

(4) 250 sec.

31) The decomposition of azomethane at certain temperature according to the equation $(CH_3)_2N_2 \rightarrow C_2H_6 + N_2$ is a first order reaction. After 40 minutes from the start the total pressure developed is found to be 350 mm of Hg in place on initial pressure 200 mm of Hg of azomethane. The value of rate constant K is :-

(1) $2.88 \times 10^{-4}\ sec^{-1}$

(2) $1.25 \times 10^{-4}\ sec^{-1}$

(3) $5.77 \times 10^{-4}\ sec^{-1}$

(4) None of them

32) For a hypothetical reaction $aA + bB \rightarrow$ Product, the rate law is, rate (r) = $k[A]^x[B]^y$ then which of the following relation is possible –

(1) a + b = x + y

(2) a + b > x + y

(3) a + b < x + y

(4) All of the above

33) In a reaction, the threshold energy is equal to :-

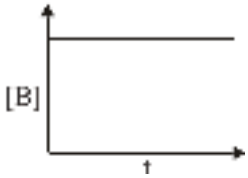
(1) activation energy + energy of reactants

(2) activation energy – energy of reactants

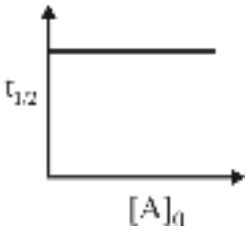
(3) activation energy

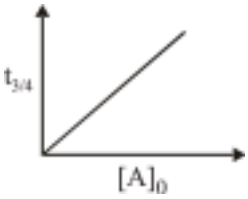
(4) energy of reactants

34) Which graph represents zero order reaction ; A(g) → B(g) :-

(1) 

(2) 

(3) 

(4) 

35) A rise of 10°C can cause the rate of some reactions to double. This is best explained by :-

(1) The average velocity of the molecules has doubled.

(2) The number of molecules with more than enough energy to overcome the activation energy barrier has doubled.

(3) The activation energy has lowered.

(4) The average kinetic energy of the particles has doubled

36) Number of σ and π bonds in propane -1, 2, 3-tricarboxylic acid are :-

(1) 16σ, 6π

(2) 19σ, 3π

(3) 21σ, 6π

(4) 18σ, 4π

37) Correct order of functional groups according to priority table of IUPAC

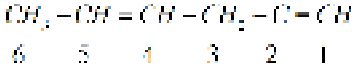
(1) –COOH > – $CONH_2$ > – COX > – OH

(2) –COX > – $CONH_2$ > – CHO > – COOR

(3) –CHO > – NH_2 > –OH > –CN

(4) –COX > –CN > –CHO > –OH

38) In the hydrocarbon :



The state of hybridization of carbons 1, 3 and 5 are in the following sequence :-

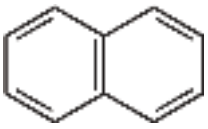
(1) sp, sp^2, sp^3

(2) sp^3, sp^2, sp

(3) sp^2, sp, sp^3

(4) sp, sp^3, sp^2

39) No. of 2° carbon are present in given compound:-



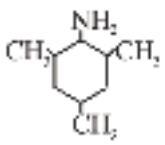
(1) 6

(2) 10

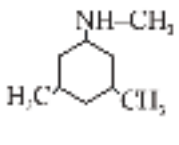
(3) 8

(4) 2

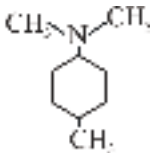
40)



(I)



(II)



(III)

The amine I and II, III are respectively :-

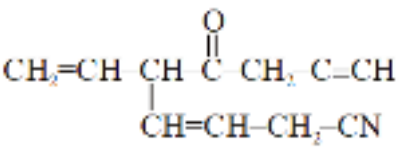
(1) 3°, 2° and 1°

(2) 1°, 2° and 3°

(3) 2°, 2° and 3°

(4) 1°, 3° and 2°

41) Number of olefinic bonds and triple bonds in the following compound respectively :



(1) 2, 2

(2) 3, 1

(3) 3, 2

(4) 2, 1

42) **Assertion :-** Isopropyl alcohol is a secondary alcohol.
Reason :- –OH attached with 2° carbon.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

43) Match the list-I and list-II and select the correct answer :-

List - I		**List - II**	
(a)	C_nH_{2n}	(p)	Alcohol, ether
(b)	C_nH_{2n-2}	(q)	Alkene, cycloalkane
(c)	$C_nH_{2n+2}O$	(r)	Aldehyde, Ketones
(d)	$C_nH_{2n}O$	(s)	Alkynes, Alkadiene

(1) (a - p), (b - q), (c - r), (d - s)

(2) (a - s), (b - r), (c - q), (d - p)

(3) (a - q), (b - s), (c - p), (d - r)

(4) (a - r), (b - p), (c - s), (d - q)

44) The IUPAC nomenclature of
$(CH_3)_3C–CH=C(CH_3)_2$ is :-

(1) 2,4,4–Trimethylpent-3-ene

(2) 2,4,4–Trimethylpent-2-ene

(3) 2,2,4–Trimethylpent-3-ene

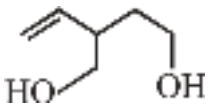
(4) 2,2,4–Trimethylpent-2-ene

45) Which is correctly written IUPAC name

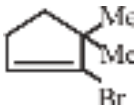
(1) 
Hepta-1, 4-dien-6-yne

(2) 

1-Bromo prop-2-ene

(3) 

2-vinyl butane-1, 4 - diol

(4) 

2-Bromo-5, 5-dimethyl cyclohex-1-ene

BIOLOGY

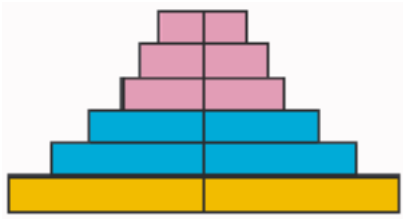
1) If '8' *Drosophila* in a laboratory population of '80' died during a week, the death rate in the population is_____individuals per *Drosophila* per week.

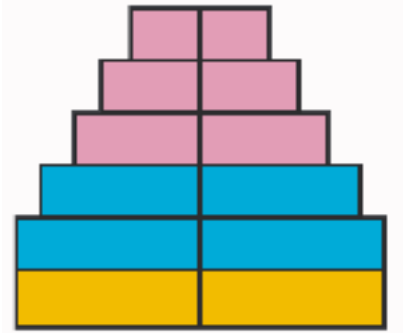
(1) 10
(2) 1.0
(3) zero
(4) 0.1

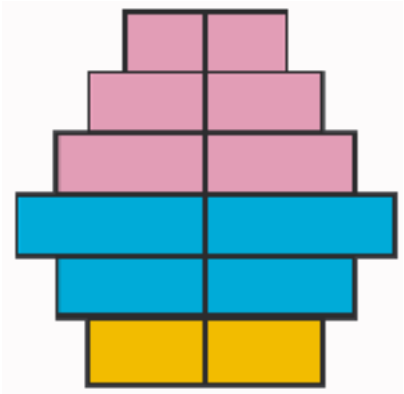
2) Darwinian fitness is represented by-

(1) Lower 'r' value
(2) High 'r' value
(3) High 'k' value
(4) Low 'k' value

3) Which one of the following age pyramid shows more steeper growth.

(1) 

(2) 

(3) 

(4) Both 1 and 3

4) In the euqation for 'S' shaped population growth $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$, in this equation K represent

(1) Carrying capacity
(2) Environmental Resistance
(3) Biotic potential
(4) Population size

5) Choose the correct option:

(1) Geometric growth produces J–shaped population growth curve
(2) Logistic growth occurs when resources are limiting
(3) For exponential growth equation is $N_t = N_0e^{rt}$
(4) All

6) In the exponential growth equation, $N_t = N_0e^{rt}$, e represents :

(1) The base of number logarithms
(2) The base of exponential logarithms
(3) The base of natural logarithms
(4) The base of geometric logarithms

7) Which of the following is not an attribute of a population?

(1) Species interaction
(2) Sex ratio
(3) Natality
(4) Mortality

8)

Natality refers to

(1) Death rate
(2) Birth rate
(3) Number of individuals leaving the habitat
(4) Number of individuals entering a habitat

9) Which method is used for tiger census ?

(1) Biomass method
(2) Average method
(3) Optical density method
(4) Pug mark and fecal pellets method

10) Which of the following examples signfies a population ?

(1) Cormorants in a wetland
(2) Rats in an abandoned dwelling
(3) Teakwood trees in a forest
(4) All of the above

11) Tropical rain forests once covering more than 14 percent of the earth's land surface, these rain forests now cover no more than 6 percent. This loss is one of the most important cause of extinction of life forms and is because of :-

(1) Overexploitation
(2) Alien species invasions
(3) Habitat loss
(4) Co-extinction

12) World summit on sustainable development held in 2002 was called :-

(1) To reduce CO_2 emissions and global warming
(2) To assess threat posed to native species by invasive weed species
(3) For immediate steps to discontinue used of CFCs that were damaging the ozone layer
(4) For a significant reduction in the current rate of biodiversity loss at global, regional and local levels.

13) How many of the following characters are not linked with biodiversity hotspots ?
(A) They are Megadiversity zones.
(B) High degree of Endemism
(C) High species richness
(D) High intensity of competition
(E) Show minimal habitat loss.

(1) One
(2) Two
(3) Three
(4) Zero

14) **Statement-I :-** The conventional taxonomic methods are not suitable for identifying microbial species.
Statement-II :- The rate of extinction in sixth episode of extinction is less than the previous episodes of extinction.

(1) Both Statement-I & Statement-II are false
(2) Statement-I is correct but Statement-II is incorrect
(3) Statement-I is incorrect but Statement-II is correct
(4) Both Statement-I & II are correct.

15) **Statement I :** Among animals, insects are the most species-rich taxonomic group.

Statement II : Out of every 100 animals on the planet, 7 are insects.
In the light of the above statements, choose the most appropriate answer from the option given bolow:

(1) Both Statement I and Statement II are incorrect
(2) Statement I is correct but Statement II is incorrect
(3) Statement I is incorrect but Statement II is correct
(4) Both Statement I and Statement II are correct

16) Biodiversity term popularised by ?

(1) Edward Wilson
(2) Alexander von humboldt
(3) Gause
(4) Connel

17) A more conservative and scientifically sound estimate made by Robert May places the global species diversity at about _____

(1) 1.5 million
(2) 7 million
(3) 70 million
(4) 1 million

18) How many Hot-spot are there in India

(1) 3
(2) 34
(3) 90
(4) 448

19) How many examples are of sacred groves
(a) Aravali Hills of Rajasthan
(b) Western ghat region of Karnataka and Maharashtra,
(c) Chanda, Bastar and sarguja in Madhya Pradesh
(d) Khasi & Jaintia hills in Meghalaya
(e) Western ghat and Srilanka

(1) a,b,c,d
(2) a,c,d
(3) a,d,e
(4) a,b,c,d,e

20) The world summit on sustainable development held in ?

(1) 2002
(2) 1991

(3) 2015

(4) 2005

21) Which of the following statement are true ?
(a) Tilman found that plots with more species showed more year to year variation in total biomass.
(b) The IUCN red list documents the extinction of 784 species in the last 500 years.
(c) The last twenty years alone have witnessed the disappearance of 37 species.
(d) In general loss of biodiversity in a region may lead to decline in plant production.

(1) a and c are correct

(2) a and b are correct

(3) b and d are correct

(4) b and c are correct

22) 

In above diagram which community have maximum and minimum species diversity respectively

(1) Maximum = B, Minimum = D

(2) Maximum = A, Minimum = D

(3) Maximum = A, Minimum = B

(4) Maximum = C, Minimum = D

23) Which of the following characters are changed with altitude and latitude ?
(a) Temperature (b) Rain fall
(c) Vegetation (d) Species diversity
Options :

(1) only b

(2) only a and b

(3) only a, b and c

(4) a, b, c & d

24) Which of the following character is not related to biodiversity hot spot.

(1) High degree of endemism

(2) Accelerated habitat loss

(3) Lesser interspecific interaction

(4) High species richness

25) In "rivet popper hypothesis" rivet indicates.

(1) Community

(2) Biosphere

(3) Ecosystem

(4) Species

26) Extinction of stellar's sea cow is due to which of the following reasons :

(1) Habitat loss
(2) Over exploitation
(3) Alien species invasion
(4) Co-extinction

27) India has only ____A_____ of the world's Land area, its share of the global species diversity is an impressive ____B_____. Identify A and B :-

(1) A-8.1%, B-2.4%
(2) A-2.4%, B-8.1%
(3) A-3.4%, B-9.1%
(4) A-9.1%, B-3.4%

28) The value of Z for frugivorous birds in the tropical forests is :-

(1) 0.6 to 1.2
(2) 0.1 to 0.2
(3) 1.15
(4) 2.15

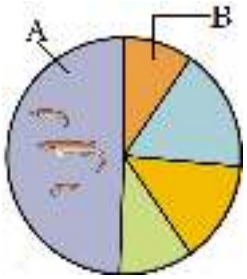
29) Identify the **incorrect** statement :-

(1) Colombia located near the equator has nearly 1400 species of bird
(2) The largely tropical Amazonian rain forest in South America has the greatest biodiversity on earth
(3) Speciation is generally a function of time
(4) 32% of all amphibian species not facing the threat of extinction

30) Match the List-I with List-II :-

List-I		**List-II**	
(A)	Habitat loss	(a)	Amazon rain forest
(B)	Over-exploitation	(b)	Clarias gariepinus
(C)	Alien species invasion	(c)	Fig tree and wasp
(D)	Co-extinction	(d)	Passenger pigeon

(1) (A)-(a), (B)-(d), (C)-(b), (D)-(c)
(2) (A)-(d), (B)-(b), (C)-(a), (D)-(c)
(3) (A)-(d), (B)-(c), (C)-(b), (D)-(a)
(4) (A)-(a), (B)-(b), (C)-(c), (D)-(d)

31) 

Identify the A and B :

(1) A-fishes, B-Reptiles
(2) A-fishes, B-Birds
(3) A-fishes, B-Mammals
(4) A-fishes, B-Amphibians

32) In the below some example are given these example related to biodiversity conservation. Biodiversity Hotspot, Biosphere reserve, national park, botanical garden, zoological park, wild life sanctuary, tissue culture. How many examples are included in Ex-situ approach.

(1) Two
(2) Five
(3) Four
(4) Three

33) Which taxonomic group is more vulnerable to extinction?

(1) Amphibians
(2) Aves
(3) Mammals
(4) Pisces

34) **Assertion:** The Amazon rain forest it is called the lungs of the planet.
Reason: The Amazon rain forest is estimated to produce 20% of the total oxygen in the earth's atmosphere.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

35) **Assertion:** India is one of the 12 mega diversity countries of the world.
Reason: India shares of the global species diversity is an impressive 8.1 percent.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

36) **Assertion:** Bioprospecting is exploring molecular, genetic and species level diversity for the product of economic importance.
Reason: More than 25 percent of the drugs currently sold in the market world wide are derived from plants.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

37) **Assertion:** Nature's biological library is burning even before we catalogue the titles of all the books stocked there.
Reason: A large fraction of species diversity is facing the threat of becoming extinct even before we discover them.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

38) **Assertion :** Western ghats and Sri Lanka, Indo-Burma and The Himalaya are considered as hotspots.
Reason : High levels of species richness and high degree of endemism is found in these regions.

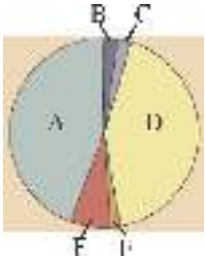
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

39) In below pie-chart of global biodiversity of vertebrates, 'A' represents birds & 'D' represents fishes then what will be 'C' & 'E' represent respectively ?



(1) Reptiles & Amphibia
(2) Mammals & Amphibia
(3) Mammals & Reptiles
(4) Amphibia & Mammals

40) Study the following pie chart with respect to the global diversity of plants and select the correct option :-



	A	B	C	D	E	F
(1)	Angiosperm	Lichen	Algae	Fungi	Mosses	Fern and allies
(2)	Fungi	Lichen	Algae	Angiosperm	Mosses	Fern and allies
(3)	Fungi	Mosses	Fern and allies	Angiosperm	Lichen	Algae
(4)	Angiosperm	Fern & allies	Lichen	Fungi	Algae	Mosses

(1) 1

(2) 2

(3) 3

(4) 4

41) In the following statements choose correct options :-
(a) Wasp laying egg in a fig fruit is an example of mutual relationship
(b) Mycorrhiza is an example of ectoparasitism
(c) Physiological ecology is ecology at organism level
(d) Competitive exclusion principle was given by connell

(1) a and c

(2) a, c, d

(3) b and d

(4) b, c and d

42) Select the **incorrect** statement from the following :-

(1) Thorns/Spine (Acacia, cactus) are the most common morphological means of defence

(2) Interspecific competition is a potent force in organic evolution

(3) Goats killed Abingdon tortoise in Galapagos island

(4) Vertical layering seen in which different species are found at different level called stratification

43) How many statements are correct ?
(a) In an aquatic ecosystem grazing food chain is major conduct for energy flow
(b) Detritus food chain may be connected with grazing food chain at some level
(c) Each tropic level has a certain mass of living material at a particular time called as standing crop
(d) Species composition and stratification are two main structural features of an ecosystem

(1) Two

(2) Three

(3) One

(4) All

44) Population density increase if :

(1) (B + I) > (D + E)
(2) (B + I) < (D + E)
(3) (B + I) = (D + E)
(4) (B + D) = (I + E)

45) Human liver fluke depends on two intermediate hosts to complete its life cycle, they are :-

(1) Snail and Frog
(2) Snake and Fish
(3) Snail and Fish
(4) Snake and Frog

46) k-selected species has :-

(1) High stability & low biotic potential
(2) Low stability & high biotic potential
(3) Short life time & high growth rate
(4) Long life time & high growth rate

47) Find out the correct match from the following table :-

(i)	Fig Tree	(a)	i, ii, v are example of key stone species
(ii)	Lion	(b)	i, ii, iii are example of key stone species
(iii)	Kangaroo rat	(c)	vi, vii, are example of critical link species
(iv)	Kiwi	(d)	iv, v, vi are example of endemic species
(v)	Kangaroo	(e)	vii is the example of critical link species
(vi)	Meta sequoia		
(vii)	Mycorrhizal fungi		

(1) a, b, c are correct
(2) a, b, c, d, e all correct
(3) b, d, e are correct
(4) a, c are correct

48) Connell's elegant field experiment exhibits :-

(1) Competitive release
(2) Co-existence

(3) Resource partitioning
(4) Exclusion principle

49) Which is a potent force in organic evolution :-

(1) Intra specific competition
(2) Inter specific competition
(3) Predation
(4) Parasitism

50) *Cuscuta* is :-

(1) Predator
(2) Partial root parasite
(3) Total stem parasite
(4) Total root parasite

51) Identify the correct match from the column I, II and III.

Column-I	**Column-II**	**Column-III**
1. Mutualism	a. (+, –)	i. Visiting flamingoes and resident fishes in south american lakes
2. Competition	b. (+, 0)	ii. Starfish pisaster in american pacific coast
3. Predation	c. (+, +)	iii. Cattle egret birds and cattle
4. Commensalism	d. (–, –)	iv. Lichens

(1) 1 - d - iv, 2 - c - i, 3 - a - ii, 4 - b - iii
(2) 1 - c - iv, 2 - d - i, 3 - a - ii, 4 - b - iii
(3) 1 - c - iv, 2 - d - iii, 3 - a - ii, 4 - b - i
(4) 1 - d - iv, 2 - c - i, 3 - b - ii, 4 - a - iii

52) Which of the following is an example of detrimental relation ?

(1) Fig and wasp
(2) Ticks and dog
(3) Mycorrhiza
(4) Lichen

53) A bird Introduced from another country become a serious pest due to :-

(1) Better Adaptation to that Area
(2) More Sexual Reproduction
(3) Better Nesting habitats

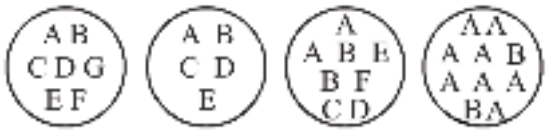
(4) Absence of Natural Predator

54) **Assertion :** Five closely related species of Warblers avoid competition by resource partitioning.
Reason : Species facing competition could avoid competition by choosing different times for feeding or different foraging patterns.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

55) **Assertion (A) :–** Predators help in maintaining the species diversity in a community.
Reason (R) :– Predators reduce the intensity of competition among the competing prey species.

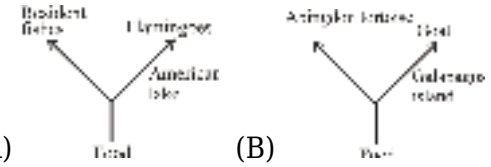
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.



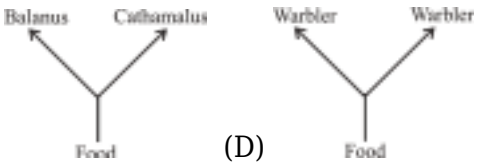
56) In above diagram four areas are given. Which area has maximum and minimum species diversity respectively.

(1) Maximum = B, Minimum = D
(2) Maximum = A, Minimum = D
(3) Maximum = A, Minimum = B
(4) Maximum = C, Minimum = D

57) (A) (B)



(C) (D)

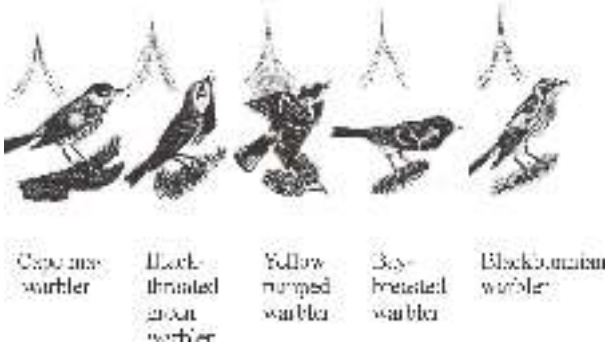


Select the correct match for above diagrams :

(1) A – Representing mutualism
(2) B – Showing Gause competitive exclusion principle
(3) C – showing resources partitioning
(4) D – showing parasitism

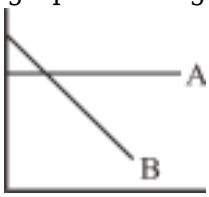
58) In the given figure the different five warblers showing interaction and this interaction is based

on which of the following principle ?



(1) Gause's competitive exclusion
(2) Interference competition
(3) Mac Arthur resource partitioning
(4) Conell's competitive reduce

59) Below given graph showing interaction between two species A and B. Identify the interaction between them ?



(1) Mutualism
(2) Commensalism
(3) Competition
(4) Amensalism

60) Which plant is pollinated by insect in figure given below ?



(1) Orchid flower
(2) Fig flower
(3) Sunflower
(4) Lotus flower

61) Study the four statements (a-d) given below and select the two correct ones out of them :
(a) A lion eating a deer and a sparrow feeding on grain are ecologically similar in being predators.
(b) Predator star fish Pisaster helps in maintaining species diversity of some invertebrates

(c) Predators ultimately lead to the extinction of prey species
(d) Production of chemicals such as nicotine, strychnine by the plants are metabolic disorders
The two correct statements are :

(1) (a) and (b)

(2) (b) and (c)

(3) (c) and (d)

(4) (a) and (d)

62) Predators help in :-
(a) Maintaining species diversity
(b) Increase the interspecific competition
(c) Control prey population
(d) Maintaining ecological balance

(1) a, b

(2) b, c

(3) a, c and d

(4) a, d

63) Which of the following are correctly matched pair?
(A) Copepods on marine fishes – Parasitism
(B) Cow grazing on grass – Predation
(C) Barnacle and whale – Commensalism
(D) Sparrow feeding on seed – Predation
(E) *Pronuba* and *Yucca* plant – Mutualism

(1) A, B, C, D and E

(2) B, C, D and E

(3) A, C, D and E

(4) A, C and E only

64) Which of the following option is true regarding predation ?

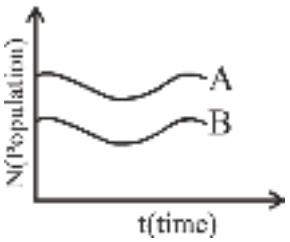
(1) In the rocky intertidal communities of the American Pacific Coast the starfish *Pisaster* is an important predator. In a field experiment, when all the starfish were removed from an enclosed intertidal area, more than 10 species of invertebrates became extinct within a year, because of intraspecific competition.

(2) The Monarch butterfly is highly distasteful to its predator (bird) because of a special chemical present in its body. Interestingly, the butterfly acquires this chemical during its caterpillar stage by feeding on a poisonous seed.

(3) A wide variety of chemical substances that we extract from plants on a commercial scale (nicotine, caffeine, quinine, strychnine, opium, etc.,) are produced by them actually as defences against grazers and browsers.

(4) All of the above

65) Which of the following is correct for the specific interaction shown in the graph ?



(1) Commensalism
(2) Mutualism
(3) Parasitism
(4) Competition

66) **Assertion :** Pyramid of energy is always upright.
Reason : When energy flows from a particular trophic level to the next trophic level, some energy is always lost as heat at each step.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

67) **Assertion :** All the steps of decomposition operate simultaneously on detritus.
Reason : Temperature & soil moisture are the two most important climatic factors that regulate decomposition.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

68) **Assertion (A) :–** Population size for any species is not a static parameter.
Reason (R) :– Population size keep changing with the time depending on various factors including food availability, predation pressure, and reduced weather.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

69) 

Above inverted pyramid of biomass also pridict that :-

(1) Small standing crop of phytoplanktans supports large standing crop of zooplankton
(2) Organism of high biotic potential supports organism of low biotic potential.
(3) Organism of high energy (productivity) supports organism of low energy (productivity)

(4) All of the above

70) Which are incorrect about ecological pyramids :

	Ecosystem	Biomass	Energy	Number
(1)	Tree			
(2)	Lake			
(3)	Grass Land			
(4)	Ocean			

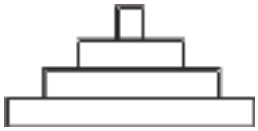
(1) 1

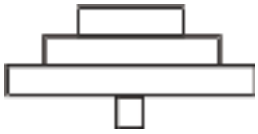
(2) 2

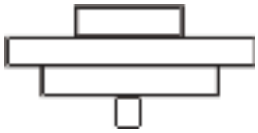
(3) 3

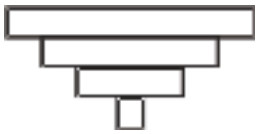
(4) 4

71) If you were to count the number of insects feeding on a big tree what kind of pyramid would you get? If the number of small birds depending on the insect, as also the number of larger birds eating the smaller:-

(1) 

(2) 

(3) 

(4) 

72) Consider the following features.
(a) Rich biodiversity
(b) Higher productivity
(c) Maximally leached soil
(d) Low standing crop
(e) Minerals stored in tall trees
(f) Seasonal variations and unpredictable environment
Select the option which holds for tropical rain forests.

(1) c, d, e & f

(2) a, b, c & e

(3) only e & f

(4) only a, b & c

73) **Statement-I :-** Pond, lake, wetland, river and estuary are some examples of aquatic ecosystems.
Statement-II :- Crop fields and an aquarium may also be considered as man made ecosystems.

(1) Statement-I and statement-II both are true
(2) Statement-I is true and statement-II is false
(3) Statement-I is false and statement-II is true
(4) Statement-I and statement-II both are false

74) **Statement-I:-** In aquatic ecosystem, DFC is major conduit of energy flow.
Statement-II:- In terrestrial ecosystem, much larger fraction of energy flows through GFC.

(1) Statement-I and statement-II both are true
(2) Statement-I is true and statement-II is false
(3) Statement-I is false and statement-II is true
(4) Statement-I and statement-II both are false

75) Decomposition is one of the important functional aspect of ecosystem. About this which of the following statements is not correct?

(1) Warm and moist environment favours decomposition.
(2) Nitrogen and sugar component favours decomposition.
(3) Low temperature and anaerobiosis favours decomposition.
(4) Decomposition is largely an oxygen requiring process.

76) Which statement is incorrect ?

(1) Trophic level represent functional level not a species as such.
(2) A same species may occupy more than one trophic level in same ecosystem at same time.
(3) In most ecosystem, all pyramids of number, or energy and biomass are upright.
(4) Energy at higher trophic level is more as compared to lower levels as they are ecologically strong.

77) Following are the structural features of an ecosystem
(a) Species composition
(b) Energy flow
(c) Decomposition
(d) Stratification

(1) (a) and (d)
(2) (b) and (c)
(3) Only (a)
(4) (b), (c) and (d)

78) Major functional aspects of ecosystem's are energy flow, dominance, stratification, decomposition, species diversity, nutrient cycle.

(1) 3

(2) 6

(3) 4

(4) 5

79) A population has certain attributes. This attributes are :

(a)	Death
(b)	Birth
(c)	Death rate
(d)	Birth rate
(e)	Sex ratio

(1) a, b, c, d, e

(2) c, d, e

(3) Only e

(4) Only c, d

80) Which of the following pair is not correctly matched ?
(A) Desert – Terrestial ecosystem
(B) Wetland – Aquatic ecosystem
(C) Pond – Open ecosystem
(D) Crop-field – Artificial ecosystem
(E) Estuary – Closed ecosystem

(1) B, C and E

(2) C and E

(3) Only E

(4) B and C

81) Omnivores is/are :
(a) Man
(b) Peacock
(c) Cockroach
(d) Crow
(e) Sparrow

(1) (a) and (b)

(2) (a), (b) and (c)

(3) (a), (b), (c) and (d)

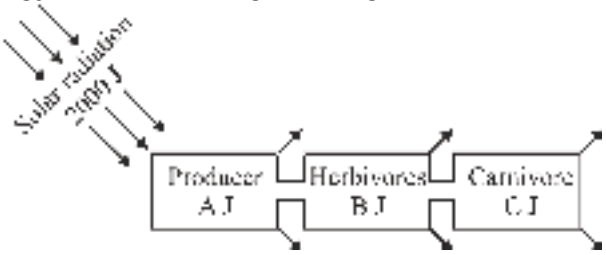
(4) (a), (b), (c), (d) and (e)

82) Identify the ecosystem based on given properties -
(a) Lack of self regulation
(b) High productivity
(c) Less complex food webs

(d) Least species diversity
(e) Little cycling of nutrients

(1) Savannah grasslands
(2) Temperate coniferous forest
(3) Tropical rain forest
(4) Agricultural croplands

83) Suppose 2000 J of solar energy is incident on green vegetation. On the basis of 10% law of Lindeman. Identify A, B and C.

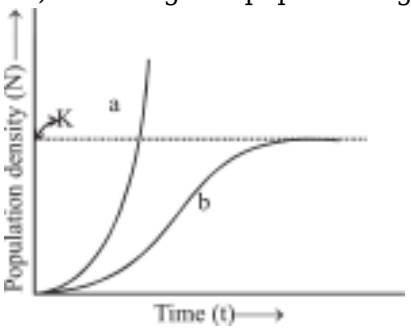


(1) A-20 J, B-2 J, C-0.2 J
(2) A-200 J, B-20 J, C-2 J
(3) A-400 J, B-40 J, C-4 J
(4) A-40 J, B-4 J, C-0.4 J

84) In a given food chain suppose the amount of energy at the fourth tropic level is 6 kJ, what will be the energy at producer level?

(1) 0.5 kJ
(2) 60 kJ
(3) 600 kJ
(4) 6000 kJ

85) For the given population growth curve which statement is **correct** :-



(1) Curve 'a' does not show saturation because rate of population growth is very low.
(2) Curve 'b' population growth rate is represented by $\frac{dN}{dt} = rN$.
(3) 'K' is carrying capacity and its value is equal for all organisms.
(4) Curve 'b' represents logistic growth rate in a population.

86) When the population density reaches the carrying capacity that stage is called as :-

(1) Exponential phase

(2) Acceleration phase

(3) Deceleration phase

(4) Asymptote phase

87) In a month of January Siberian Cranes migrate from Russia to India for breeding a survey was done -
· Till December total population of Siberian Cranes = 1200
· Number of Birth = 400
· Number of Death = 400
· Number of Cranes immigrated = 600
. Number of cranes emigrated = 600
Calculate the total population

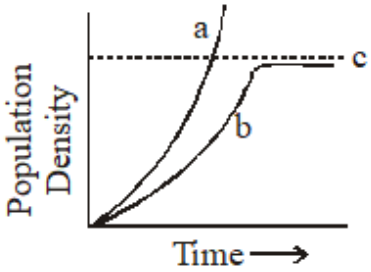
(1) 1500

(2) 1000

(3) 2000

(4) 1200

88)



(1) a = Logistic growth, b = Exponential growth, c = Carrying capacity

(2) a = Exponential growth, b = Logistic growth, c = Carrying capacity

(3) a = Carrying capacity, b = Exponential growth, c = Logistic growth

(4) a = Carrying capacity, b = Logistic growth, c = Exponential growth

89) Exploring molecular, genetic and species level diversity for product of economic importance is called :

(1) Bio remediation

(2) Bio prospecting

(3) Bio piracy

(4) Bio diversity

90) Choose the correct pair :-

(1) Oysters & pelagic fishes - breeds many time during their life time and produce small number of offsprings

(2) Pacific salmon fish - breed only once in their life time
(3) Birds and mammals - large number of small sized offsprings
(4) Bamboo - breed many times

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	3	3	2	4	1	4	1	3	4	1	4	3	1	4	1	3	3	1	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	1	1	1	1	1	2	2	3	4	2	2	2	2	4	4	2	1	2	2
Q.	41	42	43	44	45															
A.	3	2	3	2	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	2	1	3	4	3	4	4	3	1	1	4	2	3	3	3	4	4	4	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	4	3	3	3	3	2	2	1	4	3	4	1	4	2	2	4	4	3	2
Q.	86	87	88	89	90															
A.	1	1	3	2	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	2	2	1	4	3	1	2	4	4	3	4	1	2	2	1	2	1	1	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	3	4	3	4	2	2	3	4	1	3	4	1	1	1	2	1	1	4	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	3	4	1	3	1	3	1	2	3	2	2	4	1	1	2	2	3	4	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	3	1	3	2	1	2	1	1	3	2	2	1	4	3	4	1	1	2	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	4	1	4	4	4	4	2	2	2										

SOLUTIONS

PHYSICS

1) $V_M = \frac{10}{\sqrt{2}}\hat{i} + \frac{10}{\sqrt{2}}\hat{j}$

$V_W = 5\sqrt{2}\hat{i}$

Flag will be in the direction of wind with respect to moving man.

$\vec{V}_{WM} = \vec{V}_W - \vec{V}_M$

$= 5\sqrt{2}\hat{i} - \left(\frac{10}{\sqrt{2}}\hat{i} + \frac{10}{\sqrt{2}}\hat{j}\right)$

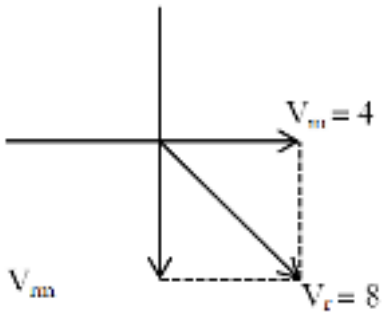
$= \frac{-10}{\sqrt{2}}\hat{j}$

Flag will flatter along south.

2) $v_{bw} + v_w = \frac{d}{t_1}$, $v_{bw} - v_w = \frac{d}{t_2}$

$2v_{bw} = d\left[\frac{1}{t_1} + \frac{1}{t_2}\right] = \frac{d}{v_{bw}} = \frac{2t_1t_2}{t_1 + t_2}$

3)



$V_{rm} = \sqrt{V_r^2 - V_m^2}$

$= \sqrt{64 - 16}$

$= \sqrt{48}$

$= 4\sqrt{3}$m/s

4)

$\Rightarrow T = \frac{2u_y}{g} \quad \Rightarrow \frac{100}{25} = \frac{2u_y}{9.8}$

$\Rightarrow u_y = 19.6$ m/s

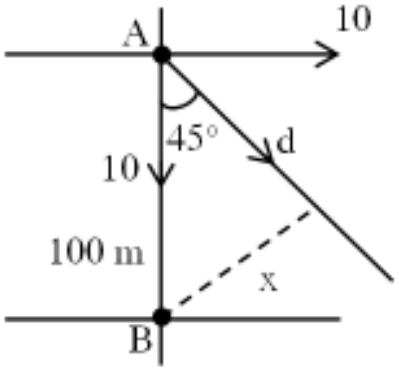
5)

$20 \sin\theta = 10 \Rightarrow \sin\theta = \frac{1}{2}$

$\theta = 30°$

$20 \cos 30° \times \frac{1}{2} = x$

$x = 10 \times \frac{\sqrt{3}}{2} = 5\sqrt{3}\,m$

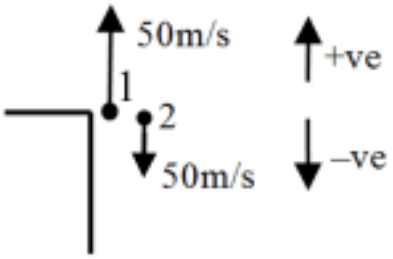


6)

$\frac{d}{100} = \cos 45°$ $\quad t = \frac{x}{V_{AB}}$

$x = 50\sqrt{2}\,m$ $\quad t = \frac{50\sqrt{2}}{10\sqrt{2}}$

$t = 5s$



7)

at any time t_1

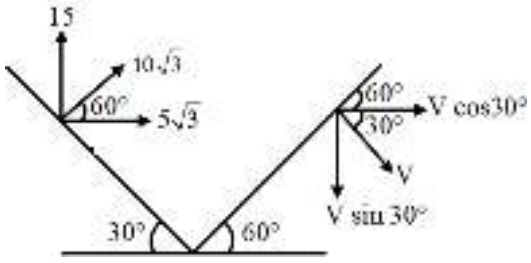
$V_1 = 50 - gt$

and $V_2 = -50 - gt$

$\vec{V}_{21} = \vec{V}_2 - \vec{V}_1$

$= -50 - gt - 50 + gt$

$\vec{V}_{21} = -100\,m/s$



8)

$V \cos 30 = 5\sqrt{3}$

$\frac{V\sqrt{3}}{2} = 5\sqrt{3}$

$V = 10$

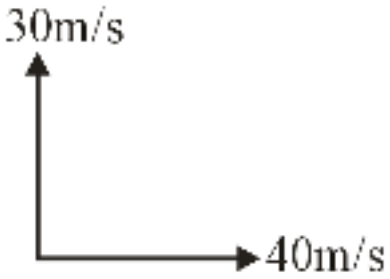
9) $y = x \tan\theta\left(1-\frac{x}{R}\right) \Rightarrow H = \frac{R}{2}\tan\theta\left(1-\frac{R}{2R}\right)$

$H = \frac{R}{4}\tan\theta$ .... (1)

$\tan 37^\circ = \frac{3}{4} = \frac{H}{\frac{R}{2}}$

$\tan\theta = \frac{4H}{R} = \frac{3}{2}$

10)



$-35 = 30t - 5t^2$

$-7 = 6t - t^2$

$t^2 - 6t - 7 = 0$

$t = -1, t = 7$

Range = $40 \times 7 = 280$ m

Man has to move = $350 - 280 = 70$m

$V_M = \frac{d}{t} = \frac{70}{7} = 10\text{m/s}$

11) $x = v_0 \cos\theta\, t$

$30 = v_0 \cos\theta \times 2$

$\Rightarrow v_0 \cos\theta = 15$

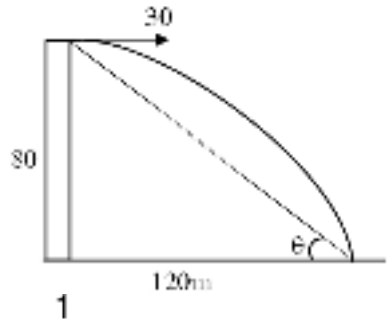
$\Rightarrow S_y = u_y t + \frac{1}{2}a_y t^2$

$\Rightarrow 44 = v^0 \sin\theta \times 2 + \frac{1}{2} \times 10 \times (2)^2$

$\Rightarrow v_0 \sin\theta = 12$

$\Rightarrow \tan\theta = \frac{12}{15} = \frac{x}{10}$ $\Rightarrow x = 8$

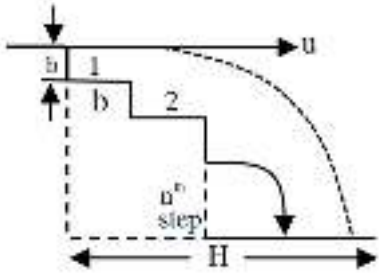
12) Ans.(4)



$h = \frac{1}{2}gt^2 = 5(4)^2 = 80$

$$\tan\theta = \frac{80}{120} = \frac{2}{3}$$

13) If the ball hits the nth step, then horizontal distance traversed = nb. Vertical distance traversed = nh.
Here, velocity along horizontal direction = u.



Velocity along vertical direction = 0.

$\Rightarrow$ nb = ut ____(i)

$$nh = 0 + \frac{1}{2}gt^2 \quad \text{____(ii)}$$

From eqn. (i),

$$t = \frac{nb}{u}, \Rightarrow nh = \frac{1}{2}g \times \left(\frac{nb}{u}\right)^2$$

$$\Rightarrow n = \frac{2hu^2}{gb^2}$$

14) $u_x = 1$; $u_y = 2$

$$R = \frac{2u_x u_y}{g} = \frac{2 \times 1 \times 2}{10} = \frac{2}{5}$$

$$\tan\theta = \frac{u_y}{u_x} = \frac{2}{1} = 2$$

$$y = x\tan\theta\left(1 - \frac{x}{R}\right)$$

$$y = x(2)\left(1 - \frac{x}{2/5}\right)$$

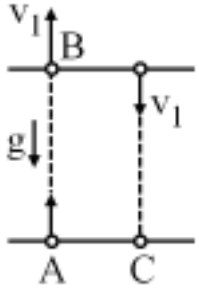
$$y = 2x\left(1 - \frac{5x}{2}\right)$$

$\Rightarrow$ $y = 2x - 5x^2$

15) For given motion
(i) Rate of change of position with time → velocity
(ii) Rate of change of velocity with time → acceleration

16)

From A to B ⇒ Velocity will decrease because gravitational acceleration is opposite direction.
From B to C ⇒ Velocity will increase because acceleration (g) is in same direction and acceleration due to gravity is always down ward.



17)

The equation of straight line $V = -\frac{1}{2}x + 2$

$\Rightarrow \frac{dv}{dx} = -\frac{1}{2}$

So, $a = V\frac{dv}{dx} = \left(-\frac{1}{2}x + 2\right)\left(-\frac{1}{2}\right)$

$\Rightarrow a = \frac{1}{4}x - 1$

So, graph between a & x is straight line shown in option 3

18) Slope of s-t graph ⇒ velocity

* Slope is constant from O to A
* Net displacement = 0, therefore average velocity = 0

19)

for accelerated motion :

$\Rightarrow v^2 = u^2 + 2as \Rightarrow v^2 = 2 \times 9.8 \times 50$

for retarded motion

$v^2 = u^2 + 2as \Rightarrow (3)^2 = 980 - 2 \times 2 \times h$

$\Rightarrow h = \frac{971}{4}$

so, total height $H = h + 50 = \frac{971}{4} + 50$

$H \simeq 293\,m$

20) Let time intervals between drops be taken by using $s = ut + \frac{1}{2}at^2$, $16 = \frac{1}{2}(10)(8T)^2$

$\Rightarrow T^2 = \frac{1}{20}$

Distance between 3rd & 5th drop

$= \frac{1}{2}(10)(6T)^2 - \frac{1}{2}(10)(4T)^2 = 100T^2$

$= 100\left(\frac{1}{20}\right) = 5m$

21) $S_x = u_xt + \frac{1}{2}a_xt^2 \Rightarrow S_x = \frac{1}{2} \times 6 \times 16 = 48$ m

$S_y = u_y t + \frac{1}{2} a_y t^2 \Rightarrow S_y = \frac{1}{2} \times 8 \times 16 = 64$ m

$S = \sqrt{S_x^2 + S_y^2} = 80$m

22) $a = V\frac{dv}{dx}$

$a = (3x^2 + 4)\ 6x$

$a = (12 + 4)\ 12$

$a = 16 \times 12 = 192\ m/s^2$

23) $x^2 = t^2 + 1$

$$\frac{dx}{dt} = \frac{t}{\sqrt{t^2+1}}$$

$$\frac{d^2x}{dt^2} = \frac{t}{\left(t^2+1\right)^{3/2}} = \frac{1}{x^3}$$

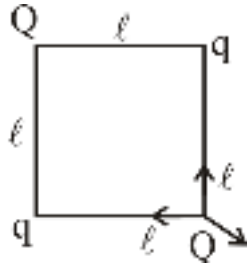
24) $$a_{avg} = \frac{60+120}{\frac{60}{10}+\frac{120}{20}} = \frac{180}{12} = 15\text{km/hr}$$

25) $\vec{d}_1 = 30\hat{j}$

$\vec{d}_2 = 20\hat{i}$

$$\vec{d}_3 = 20\sqrt{2}\left(\frac{-\hat{i}-\hat{j}}{\sqrt{2}}\right) = -20\hat{i} - 20\hat{j}$$

$d_{net} = 30\hat{j} + 20\hat{i} - 20\hat{i} - 20\hat{j} = 10\hat{j}$

26) 

⇒ If net electrical force on Q is zero.

$$\frac{kQ^2}{(\ell\sqrt{2})^2} + \sqrt{2}\frac{kQq}{\ell^2} = 0$$

Then

$$\Rightarrow \frac{Q}{2} + \sqrt{2}q = 0$$

$$\Rightarrow \frac{Q}{q} = -2\sqrt{2}$$

27) $F = ma$

$\Rightarrow \quad a = \frac{F}{m}$

$= \frac{qE}{m}$

$= \frac{5 \times 10^{-6} \times 2 \times 10^{5}}{10^{-3}}$

$= 10^3 \text{ m/s}^2$

$v^2 = u^2 - 2as$

$\Rightarrow \quad 0 = (20)^2 - 2 \times 10^3 \times s$

$\Rightarrow \quad s = \frac{400}{2 \times 10^3} = 0.2\text{m}$

28) $\vec{E} = -\frac{dV}{dr} = -\frac{\partial V}{\partial x}\hat{i} - \frac{\partial V}{\partial y}\hat{j}$

$\vec{E} = -\left[(+3x)\hat{i} - \frac{y}{2}\hat{j}\right] = (-3\hat{i} + \hat{j})$

29) $E = \frac{10}{0.1 \sin 30^\circ} = 200$ V/m

30)

$W = q(V_Q - V_P)$

$= -100e\ (-4 - 10)$

$= 14 \times 1.6 \times 10^{-17}$

$= 2.24 \times 10^{-16}$ J

31)

Assertion (A) : "The acceleration of electron and proton are different in same field".
This statement is "True" Because,

$a = \frac{F}{m} = \frac{qE}{m}$

for electron $a_{electron} = \frac{eE}{m_e}$.....(i)

& for proton $a_{proton} = \frac{eE}{m_p}$ .....(ii)

Clearly, $a_{electron} \neq a_{proton}$

Reason (R): "Electric force acting on unit positive charge is independent of mass".
This statement is also "True". Because, $F_{elec} = qE$
But "R" doesn't explain "A".

32)

$V^0 = \frac{kq_1}{a} + \frac{kq_2}{b} + \frac{kq_3}{c}$

$q_1 = \sigma.4\pi a^2$, $q_2 = -\sigma 4\pi b^2$, $q_3 = \sigma 4\pi c^2$

$k = \frac{1}{4\pi\varepsilon_0} \Rightarrow V^0 = \frac{1}{4\pi\varepsilon_0}(\sigma 4\pi a - \sigma 4\pi b + \sigma 4\pi c)$

$V^0 = \frac{\sigma}{\varepsilon_0}(a - b + c)$

33) $\Delta V = -K\lambda \ell n\left(\frac{r_B}{r_A}\right) = -K\lambda \ell n2$

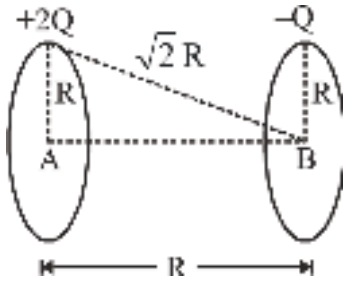
$W = q\Delta V = -k\lambda q \ell n2$

34) $V_0 = \frac{KQ}{R} V_A = \frac{KQ}{6R}$

$w_{ext} = -Q[V_0 - V_A]$

$= -Q\left[\frac{KQ}{R} - \frac{KQ}{6R}\right] = -\frac{5KQ^2}{6R}$

35)



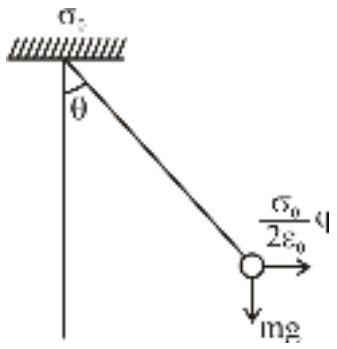
$V_A = \frac{1}{4\pi\in_0}\left[\frac{2Q}{R} + \frac{-Q}{\sqrt{2}R}\right] = \frac{Q(2\sqrt{2}-1)}{4\pi\in_0\sqrt{2}R}$

$V_B = \frac{1}{4\pi\in_0}\left[\frac{-Q}{R} + \frac{2Q}{\sqrt{2}R}\right] = \frac{Q}{4\pi\in_0 R}[(\sqrt{2}-1]$

$= \frac{Q}{4\pi\in_0\sqrt{2}R}[(2-\sqrt{2}]$

$W_{B\to A} = q(V_A - V_B) = \frac{qQ}{4\sqrt{2}\pi\in_0 R}[3\sqrt{2}-3]$

36)



$\tan\theta = \frac{\sigma q}{2\varepsilon_0 mg}$

$\frac{\tan\theta_1}{\tan\theta_2} = \frac{\sigma_1}{\sigma_2} = \frac{1}{\sqrt{3}\sqrt{3}}$

$$\frac{\sigma_1}{\sigma_2} = \frac{1}{3}$$

37) $$E_{ring} = \frac{kqx}{(R^2 + x^2)^{3/2}}$$

$$E_{Non-cond.} = \frac{kqx}{R^3} \ (R < R) = \frac{kq}{R^2} \ (r = R)$$

$$= \frac{kq}{r^2} \ (r > R) \quad E_{shell} = 0 \quad (r < R) = \frac{kq}{r^2} \ (r = R)$$

$$= \frac{kq}{r^2} \ (r > R)$$

38) 

Given: $\lambda = a(2L - x)^2$

Field at P due to small element dx -

$$dE = \frac{K(\lambda dx)}{x_1^2} = \frac{K\alpha(2L-x)^2 dx}{x_1^2}$$

Net field at P -

$$E = K\alpha \int_O^L \frac{(2L-x)^2}{x_1^2} = K\alpha \int_O^L \frac{(2L-x)^2}{(2L-x)^2} dx$$

$$E = K\alpha \int_O^L dx = K\alpha L$$

39)

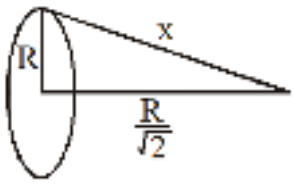
Electric field due to uniformly charged arc $E = \frac{2K\lambda}{r} \sin\frac{\phi}{2}$

$$\frac{2K\lambda}{r} \sin\frac{\phi}{2} = \frac{\sqrt{3}\lambda}{4\pi\in_0 r}$$

$$\frac{2\lambda}{4\pi\in_0 r} \sin\frac{\phi}{2} = \frac{\sqrt{3}\lambda}{4\pi\in_0 r}$$

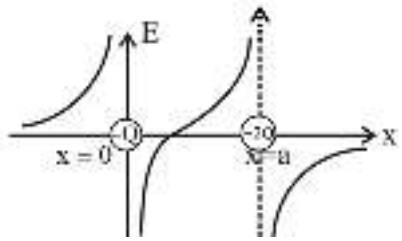
$$\sin\frac{\phi}{2} = \frac{\sqrt{3}}{2} = \sin 60° \Rightarrow \frac{\phi}{2} = 60°$$

$\phi = 120°$

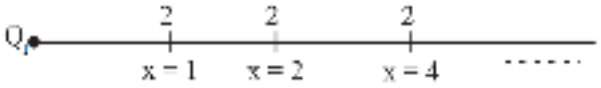
40) 

$x = \sqrt{R^2 + \frac{R^2}{2}} = \sqrt{\frac{3}{2}}R$

41)



42)



$Q_1 = 1\ C$

$Q_2 = 1\ \mu C$

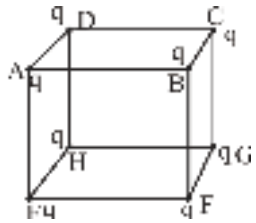
$$F = \frac{KQ_1Q_2}{1^2} + \frac{KQ_1Q_2}{2^2} + \frac{KQ_1Q_2}{4^2} + ...$$

$$F = KQ_1Q_2\left(\frac{1}{1-\frac{1}{4}}\right) = \frac{9\times10^9\times10^{-6}}{\left(\frac{3}{4}\right)} = 12000\,N$$

43)

X - Ray will ionize the air around the leaves into positive and negative charges. Hence, the charge created in the air opposite to the gold leaf will move towards it. the opposite charge neutralize the charge on gold leaf.

44)



By symmetry ϕ due to charge at E, F, G & H from face ABCD are equal

ϕ due to charge at $= \left(\frac{q}{8}\right) \times \frac{1}{\epsilon_0} \times \frac{1}{3}$ (q_{in}, 3 face)

ϕ due to charge at E, F, G, H = $4 \times \frac{q}{24\epsilon_0} = \frac{q}{6\epsilon_0}$

ϕ due to charge at A, B, C & D = 0 field lines are perpendicular to area vector

45)

$$F_1 = \frac{kq_1q_2}{(\ell \sin 30^\circ)^2}$$

$$F_2 = \frac{kq_1q_3}{(\ell \cos 30^\circ)^2}$$

So $\frac{F_1}{F_2} = \frac{q_2(\cos 30^\circ)^2}{q_3(\sin 30^\circ)^2} = \frac{3}{2} \times \left(\sqrt{3}\right)^2 = \frac{9}{2}$

CHEMISTRY

46)

INA = z axis
$p_z - p_z$ = σ bond
$p_x - p_x$ = π bond
Or
$p_y - p_y$ = π bond

47) In P → 3pπ – 3pπ is weak

48)

X = 3, Y = 1, Z = 2, W = 0

49)

PE ↓ stability ↑

50)

Max C.N of 2nd period element is 4
so CF_6^{2-} and BF_6^{-3} do not exist
F_3^- do not exist because F do not have vacant orbital.

51) **Explaion Question:** Question is asking about incorrect option regarding Lewis dot structure
Concept: Lewis dot structure, Formal charge, Resonance
Solution: The lowest energy structure is one with the lowest formal charge on the atoms.
Final Answer: 3

52)

B_2H_6 & $AlCl_3$ are
e^- defficient

53) CO_3^{2-} ⇒ octet is complete

54)

ΔH_{eg} of noble gas max.
positive in a period.

55)

Acidic strength ∝ O.S. of oxides

56) **Answer -** Option(1)
Explanation - Let's break down each statement to determine the correct answer
Concept - The effective nuclear charge is the net positive charge experienced by an electron in an atom. It's influenced by the number of protons in the nucleus and the shielding effect of inner electrons.

A. Na^+ has 11 protons and 10 electrons.
B. Al^{3+} has 13 protons and 10 electrons.

Since Al^{3+} has more protons for the same number of electrons, the effective nuclear charge is greater. This stronger pull from the nucleus contracts the electron cloud, resulting in a smaller ionic radius for Al^{3+} compared to Na^+. Therefore, Statement 2 *correctly explains* Statement 1.

57)

EA of < EA of < IP of < IP of F
F Cl Cl

58)

no. of Ve^- = large jump between successive IP.

59) **Explanation :** Max IE is asked
Concept : +ve charge has more IE than it's respactive atom,
Solution/Explanation/Calculation :
$Ne^{\oplus} > Ne$
↗
Although 'Ne' has stable e^- configuration but +ve charge has max IE than it's respactive atom because Z_{eff} ↑
Hence, the correct answer is option (3).

60) IE of Ga > IE of Al

61) ∴ $T = t_{1/2} \times n$

∴ $n = \frac{T}{t_{1/2}} = \frac{12}{3} = 4$

$N = N_0\left(\frac{1}{2}\right)^n \Rightarrow \frac{N}{N_0} = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$

62)

Explaining Question:
Find rate of disappearance of O_2 given SO_3 forms at 100 g/min.
Concept: This question is based on Stoichiometry & Rate relation
Solution:
Reaction: $2SO_2 + O_2 \rightarrow 2SO_3$
SO_3 forms at 100 g/min → corresponds to 2 mol parts
So O_2 reacts in 1 mol part
 Rate of O_2 disappearance = (1/2) × 100 = 50 g/min
BUT O_2 molar mass = 32, SO_3 = 80
So actual mol rate:
100 g SO_3/min ÷ 80 = 1.25 mol/min
O_2 disappears at 1/2 × 1.25 = 0.625 mol/min
Mass of O_2 disappearing = 0.625 × 32 = 20 g/min

63)

$C^{(1-n)} - C_0^{(1-n)} = (n - 1)\,kt$
put n = 3 $\quad C^{-2} - C_0^{-2} = 2\,kt$
concentration i.e. = $\frac{mol}{\ell}$
$\left(\frac{mol}{\ell}\right)^{-2}$ = K × second
k = ^{2} mol^{-2} sec^{-1}

64) $t_{1/2} = \frac{0.693}{K}$, $\quad 60 = \frac{0.693}{K}$
$K = \frac{0.693}{60}$ min^{-1} $\quad Kt = 2.303 \log \frac{a_0}{a_0 - x}$
$\frac{0.693}{60} \times t = 2.303 \log \frac{8}{7}$
t = 11.56 minutes

65) $E_{af} - E_{ab} = \Delta H$
$70 - E_{ab} = -22$
E_{ab} = 92 kCal

66)

Catalyst decreases Ea, Hence Increasing rate of Rxn.

67)

(i) r = K (zero order)
(ii) T↑ ROR↑
(iii) T↑ K↑
(iv) K depends on temp & catalyst.

68) $t_{1/2} \propto \frac{1}{a^{1/2}}$
n – 1 = 1/2
n = 1 + 1/2 = 3/2

69) $K_1 = K_2$

70)

Activation energy depends only on catalyst Not on stoichiometric coefficients.

71) $\Delta P = -\frac{[1.2-2]}{50-0}$
$\Delta P = \frac{0.8}{50 \times 60} \frac{atm}{sec}$
$\frac{\Delta P}{\Delta t} = \frac{\Delta C}{\Delta t} RT$
$\frac{\Delta C}{\Delta t} = \frac{0.8}{50 \times 60} \times \frac{1}{0.0821 \times 298}$
$\frac{\Delta C}{\Delta t} = 1.089 \times 10^{-5}$ mol/Lsec

72) ΔC[(Ester)] = 0.17–0.31
= –0.14 mol/L
ΔT = 60-30
= 30 sec
Avg Rate = $\frac{-[-0.14]}{30}$
$= \frac{0.14}{30}$
$= \frac{0.014}{3}$
$=4.67 \times 10^{-3}$ mol/ L sec

73)

$2A \rightarrow 3B$
t = 0 5 0
t = 10 min 5–2x 3x
$n_B = 3x = 6$
x = 2
5 – 2x = 5 – 4 = 1 mole
ROD[A] = $\frac{-[1-5]}{10-0} = \frac{4}{10} \frac{mol}{min}$

74)

For first order reaction.

$$k = \frac{2.303}{t}\log\frac{100}{100-99}$$

$$\frac{0.693}{6.93} = \frac{2.303}{t}\log\frac{100}{1}$$

$$\frac{0.693}{6.93} = \frac{2.303 \times 2}{t}$$

t = 46.06 min

75) $r = k;\ t_{1/2} = \frac{A_0}{2k}$

76)

	$(CH_3)_2N_2 \rightarrow C_2H_6 + N_2$		
t = 0	200 mm	0	0
t = 40 min	200 – P'	P'	P'

200 + P' = 350 ⇒ P' = 150 mm

$$K = \frac{2.303}{t}\log\frac{P_0}{P_0 - P'}$$

$$\Rightarrow K = \frac{2.303}{40 \times 60}\log\frac{200}{50}$$

$$= \frac{2 \times 0.693}{2400} = 5.77 \times 10^{-4}\ sec^{-1}$$

77) Order of Rxn may or may not be Equal to stichiometric coefficients
Order = x + y
sc = a + b
x + y > a + b
x + y < a + b
x + y = a + b

78) **Relalevent formulas :**
Threshold energy = Activation
Energy + Energy of reactants

79) $t_{3/4} = \frac{[A]_0 - \frac{[A]_0}{4}}{K}$

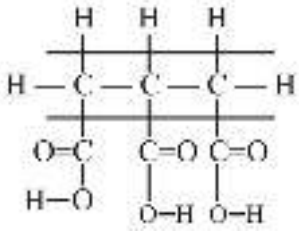
$$t_{3/4} = \frac{3[A]_0}{4K}$$

$t_{3/4} \propto [A]_0$

80)

Fact

81)

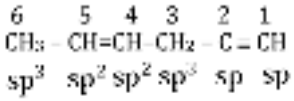


19σ, 3π

82)

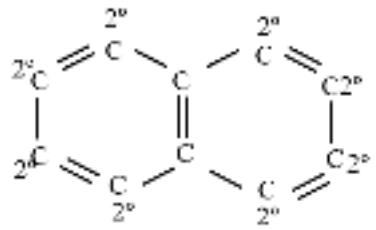
According to priority table of F.G.

83)



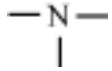
84)

Question Asking About: Concept and problems Approach
Number of 2º (secondary) carbon in given structure
Key Concept:
Carbon atom attached to two carbon is called as 2º/secondary carbon.
Solution/Explanation/Calculation:



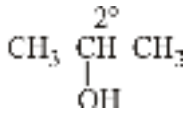
 Number of 2º carbon are :- 8
Conclusion:
Hence, option (3) is correct.

85)

1° ⇒ $-NH_2$ group
2° ⇒ –NH– group
3° ⇒  group

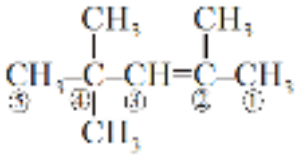
86)

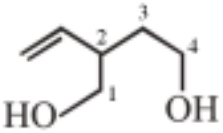
Olefinic bond ⇒ 2
Triple bonds ⇒ 2

87) 
isopropyl alcohol

88)

Compound	General formula
(i) Alcohol, Ether	$C_nH_{2n+2}O$
(ii) Alkene, Cycloalkane	C_nH_{2n}
(iii) Aldehyde, Ketone	$C_nH_{2n}O$
(iv) Alkyne, Alkadiene	C_nH_{2n-2}

89)  2, 4,4-Trimethylpent-2-ene

90) 
2-vinyl butane-1,4-diol

BIOLOGY

91)

NCERT-XII_Pg. # 191

92)

NCERT-XII_Pg. # 194

93)

NCERT-XIIth Pg.# 192

94)

NCERT-XII_Pg. # 194

95)

NCERT-XII_Pg. # 194

96)

NCERT XII Pg # 194

97)

NCERT-XII_Pg. # 191

98)

NCERT-XII, Pg. # 228

99) NCERT Pg. # 228

100) NCERT Page # 227

101) NCERT-XII, Pg # 222 & 223

102) NCERT-XII, Pg # 267

103)

NCERT XII Pg # 224

104)

NCERT - XII, Pg.# 260, 264

105)

NCERT-XII_Pg. # 218

106) NCERT-XII_Pg. # 216

107) NCERT Page No. 259

108) NCERT Page No. 266

109) NCERT Page No. 267

110) NCERT Page no. 225

111)

NCERT-XII_Pg. # 217

112)

Module-8

113)

Module-8

114)

NCERT-XII_Pg. # 224

115)

NCERT-XII_Pg. # 221

116)

NCERT-XII_Pg. # 223

117) NCERT-XII_Pg. # 217

118) NCERT Pg. # 261

119) NCERT Pg. # 264

120) NCERT Pg. # 265

121) NCERT Pg. # 260

122) NCERT Pg. # 267

123)

NCERT Pg. # 222

124) NCERT XII, Pg. # 224, 222

125) NCERT-Page No. 261

126) NCERT-Page No. 265

127) NCERT Pg. # 261, 287 (4)

128) NCERT-XII, Pg. # 266

129) NCERT XII Pg. # 260, (Fig. 15.1)

130) NCERT-XII Page # 218

131)

NCERT-XII_Pg. # 202

132)

NCERT-XII_Pg. # 197

133)

NCERT XII, Pg. # 209-210

134) NCERT, Pg # 193

135) NCERT XIIth Pg # 200

136)

Module-8

137)

NCERT XII, Pg. # 202

138)

NCERT XII, Pg. # 234

139)

NCERT, Pg. # 199

140)

NCERT XII, Pg. # 200

141)

NCERT-XII, Pg # 197-200

142)

NCERT-XII_Pg. # 197

143)

NCERT-XII_Pg. # 198

144) NCERT Pg. # 235

145) NCERT XII, Pg. # 233

146)

Module - 8

147) NCERT XIIth Pg # 234,235

148)

NCERT, Pg. # 199

149)

NCERT-XII_Pg. # 197

150)

NCERT XII, Pg # 202

151) **Question Explanation:**
Study the four statements (a-d) given below and select the two correct ones out of them :

(a) A lion eating a deer and a sparrow feeding on grain are ecologically similar in being consumers
(b) Predator star fish Pisaster helps in maintaining species diversity of some invertebrates
(c) Predators ultimately lead to the extinction of prey species
(d) Production of chemicals such as nicotine, strychnine by the plants are metabolic disorders
The two correct statements are :

Solution:

A. (a) A lion eating a deer and a sparrow feeding on grain are both consumers, as they rely on other organisms for food. This is ecologically correct.
B. (b) The predator starfish Pisaster helps maintain species diversity in some invertebrates by preying on mussels, which allows other species to thrive. This is true and demonstrates the concept of keystone species.

Hence,
Correct Option: 1

152)

NCERT-XII_Pg. # 197

153)

NCERT-XII_Pg. # 197

154) New NCERT Pg # 216, 217

155)

Module - 8

156)

Explaining the Question :-
This question also asks about the upright nature of the energy pyramid and the reason behind it.
Concept :- Pyramid of energy
Solution :-
Assertion : Pyramid of energy is always upright
Reason : When energy flows from a particular trophic level to the next trophic level, some energy is always lost as heat at each step.
As energy flows up, a significant portion (about 90%) is lost as heat at each level. Therefore, less energy is available to higher trophic levels → always upright shape.
Final Answer :-

157)

Explaining the Question :-

This question asks whether all steps of decomposition happen simultaneously and if temperature and soil moisture regulate decomposition.
Concept :- decomposition
Solution :- assertion : Decomposition involves fragmentation, catabolism, humification, and mineralization acting together on detritus.
reason : Environmental factors like temperature and moisture greatly influence the rate of decomposition by affecting microbial activity.
The reason does not fully explain why all steps operate simultaneously, so the reason is not the direct explanation for the assertion.
Final Answer :- (2)

158) NCERT XII, Pg. # 228

159)

NCERT Pg. # 248

160)

NCERT XII, Pg. # 212

161) NCERT XII, Pg. # 249

162)

Module - 8

163)

NCERT Pg.# 241

164)

NCERT Pg.# 246

165)

NCERT-XII_Pg. # 207

166)

NCERT-XII_Pg. # 209

167)

NCERT Pg. # 242

168)

NCERT-XII_Pg. # 206

169) NCERT Page No. # 229

170)

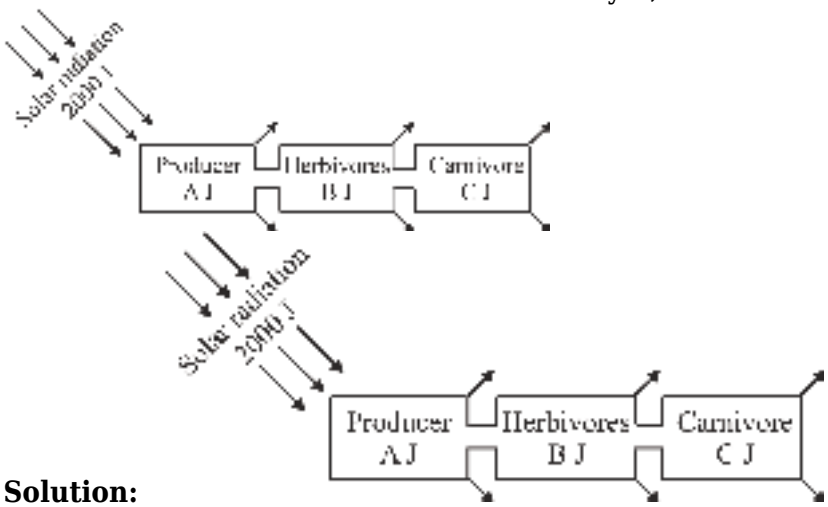
NCERT-XII, Pg. # 205 (E), 224 (H)

171) NCERT, Pg # 210

172)

NCERT-XII_Pg. # 205

173) **Question Explanation:** Suppose 2000 J of solar energy is incident on green vegetation. On the basis of 10% law of Lindeman. Identify A, B and C.



Solution:
2000 J of solar energy is incident on green vegetation. On the basis of 10% law of Lindeman. A-20 J, B-2 J, C-0.2 J

Hence,
Correct Option: 1

174)

NCERT-XII_Pg. # 213

175)

NCERT-XII, Pg.# 194, 195

176) NCERT XII, Page # 230

177)

NCERT-XII_Pg. # 193

178)

NCERT-XII_Pg. # 194

179)

NCERT-XII_Pg. # 223

180)

NCERT-XII_Pg. # 196